VICTORY 2027



Climate

LECTURE-5 Geography

Sameer Sir

Topics To Be Covered

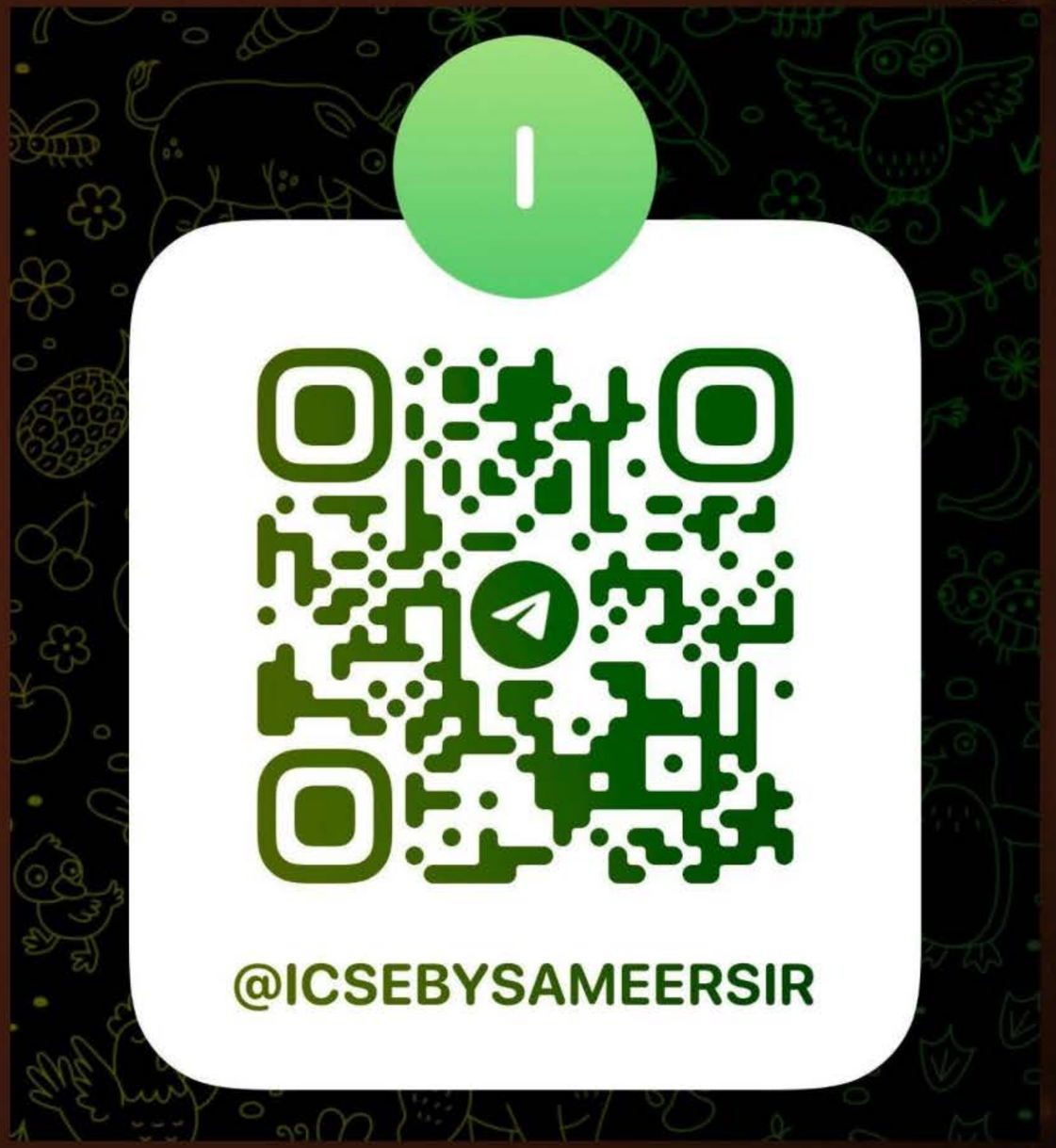
1. Revision
2.
3.
4.

Climate

Distribution of Temperature, Rainfall, winds in Summer and Winter and factors affecting the climate of India (latitudes, altitude, distance from sea, winds, Himalayas, jet streams, and El Nino). Monsoon and its mechanism. Seasons: Summer; Monsoon, Retreating Monsoon and Winter. Temperature and pressure conditions, wind movements, local winds, precipitation and interpretation of climate data.

Telegram

https://t.me/icsebysameersir

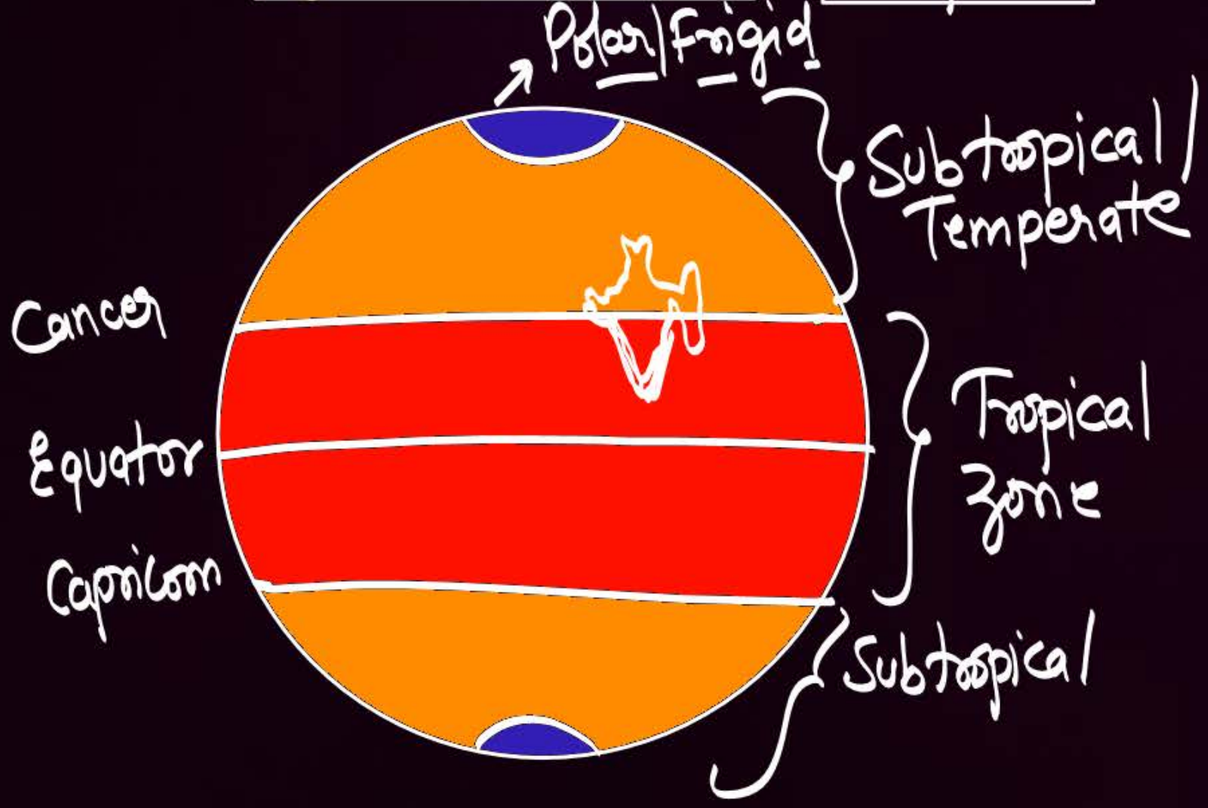


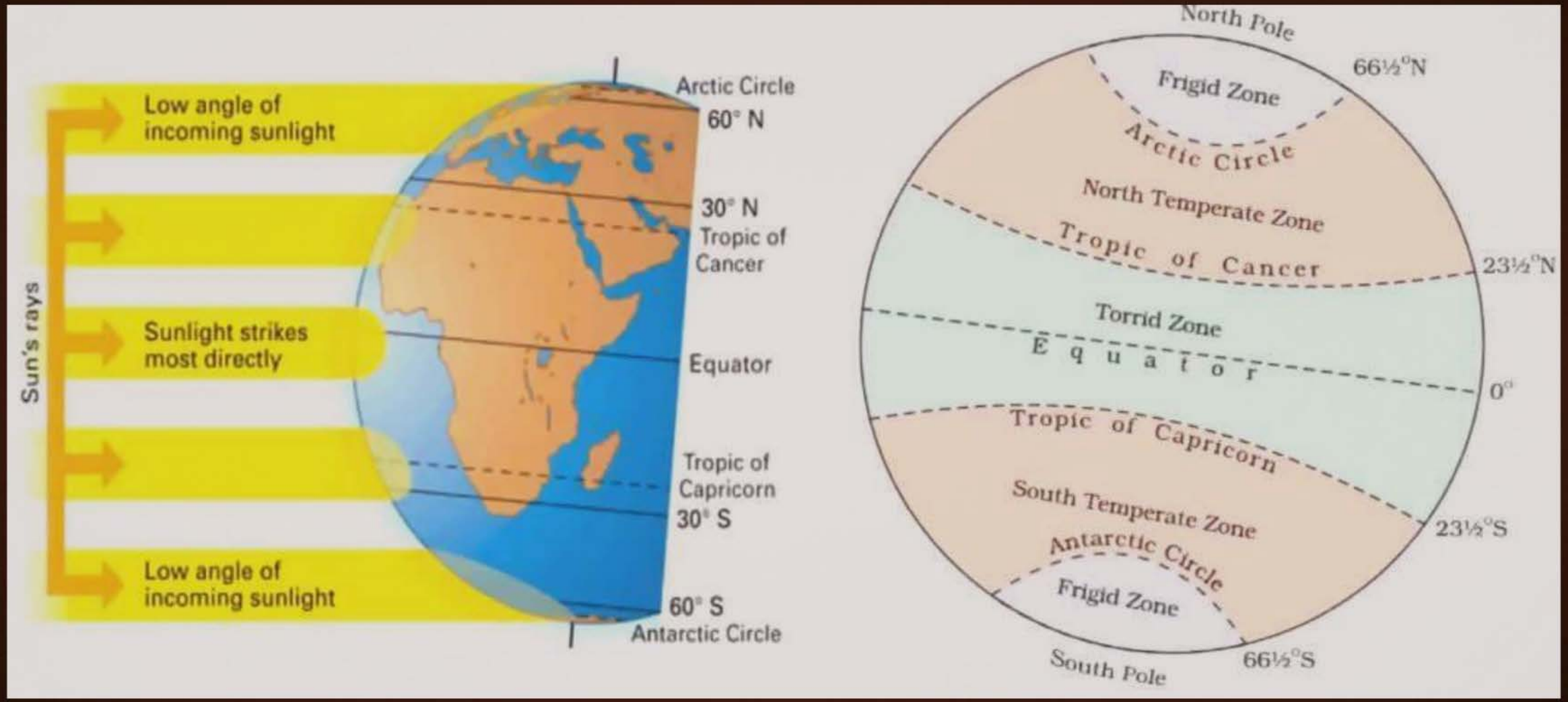
Indian Climate → Tropical Monsoon Climate

↳ Arabic word → Mausim

↓

Seasonal reversal of Winds




Low angle of incoming sunlight
Sunlight strikes most directly
Low angle of incoming sunlight
Sun's rays
Arctic Circle
60° N
30° N
Tropic of Cancer
Equator
Tropic of Capricorn
30° S
60° S
Antarctic Circle
North Pole
66½°N
Frigid Zone
Arctic Circle
North Temperate Zone
Tropic of Cancer
23½°N
Torrid Zone
Equator
0°
Tropic of Capricorn
South Temperate Zone
23½°S
Antarctic Circle
Frigid Zone
66½°S
South Pole

Continentality → Large land mass the experiences extreme weather Conditions.

↳ Excessive { Cold / Heat

Maritime Influence → Places surrounded by sea experience moderate weather Conditions.

Land → Heats / Cools } Quickly

Water → Heats / Cools } Slowly

↓

Sea Breeze → Day

Land Breeze → Night

Breeze → Winds/ हवा

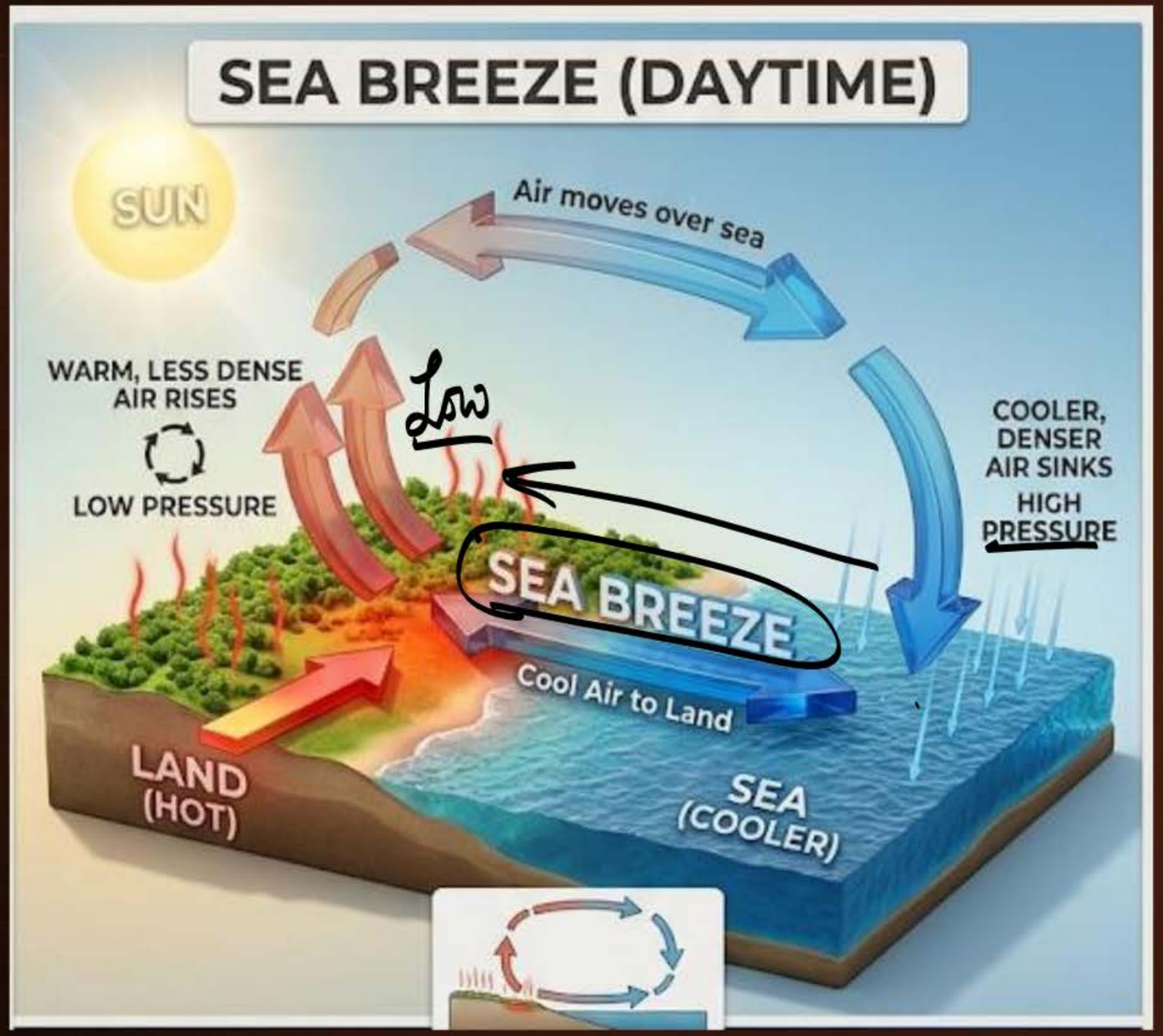
Heat → High temperature (43°C)

↓

Low pressure ← High pressure

Cold - Low temperature (27°C)

↓


SEA BREEZE (DAYTIME)
SUN
Air moves over sea
WARM, LESS DENSE AIR RISES
LOW PRESSURE
Low
COOLER, DENSER AIR SINKS
HIGH PRESSURE
SEA BREEZE
Cool Air to Land
LAND (HOT)
SEA (COOLER)


LAND BREEZE (NIGHTTIME)
MOON
Air moves over land
COOLER, DENSER AIR SINKS
HIGH PRESSURE
Cold - High
WARMER, LESS DENSE AIR RISES
LOW PRESSURE
LAND BREEZE
Cool Air to Sea
LAND (COOLER)
SEA (WARMER)

Diurnal Range → Difference between the lowest and Highest temperature of the day.

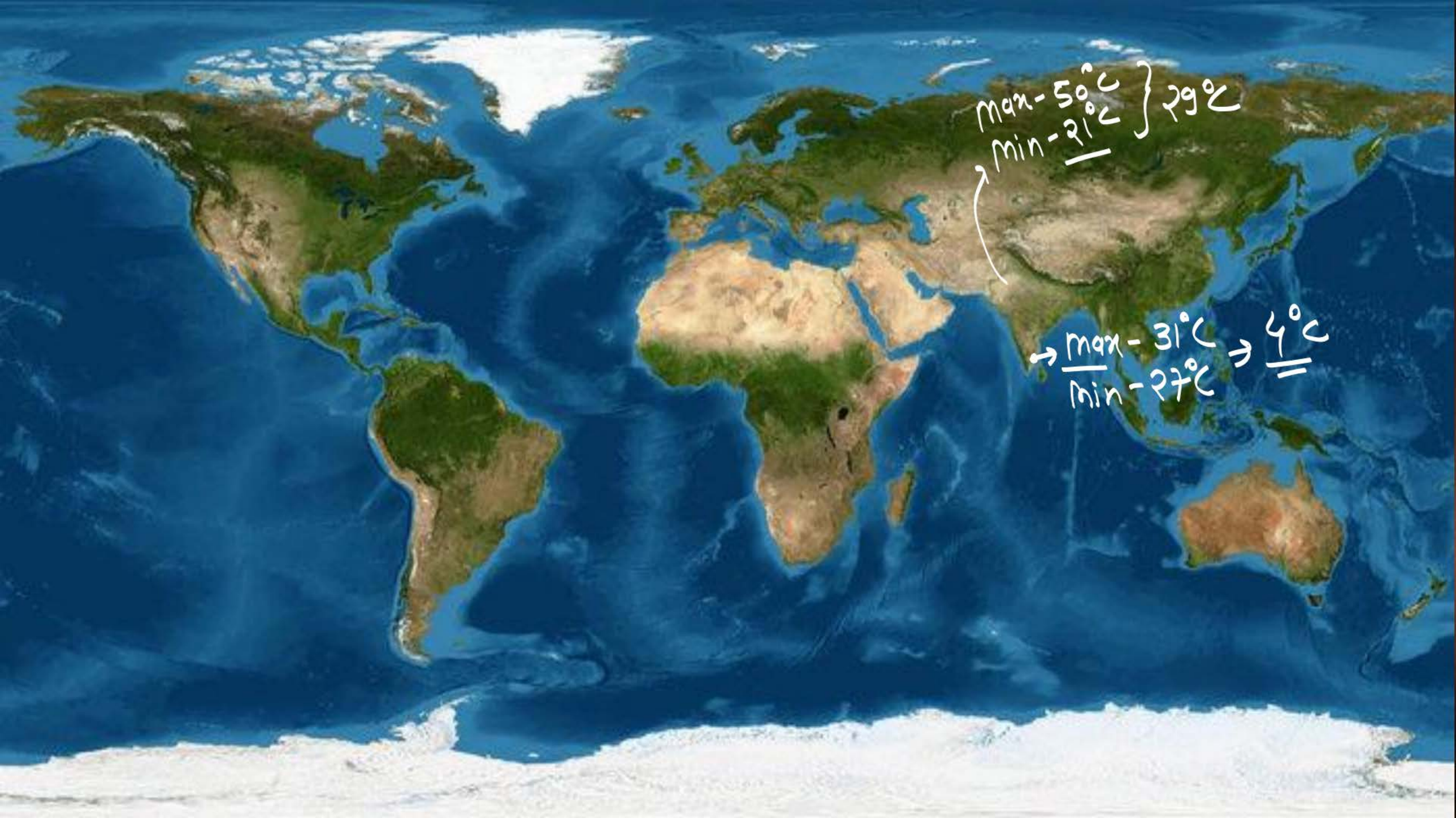
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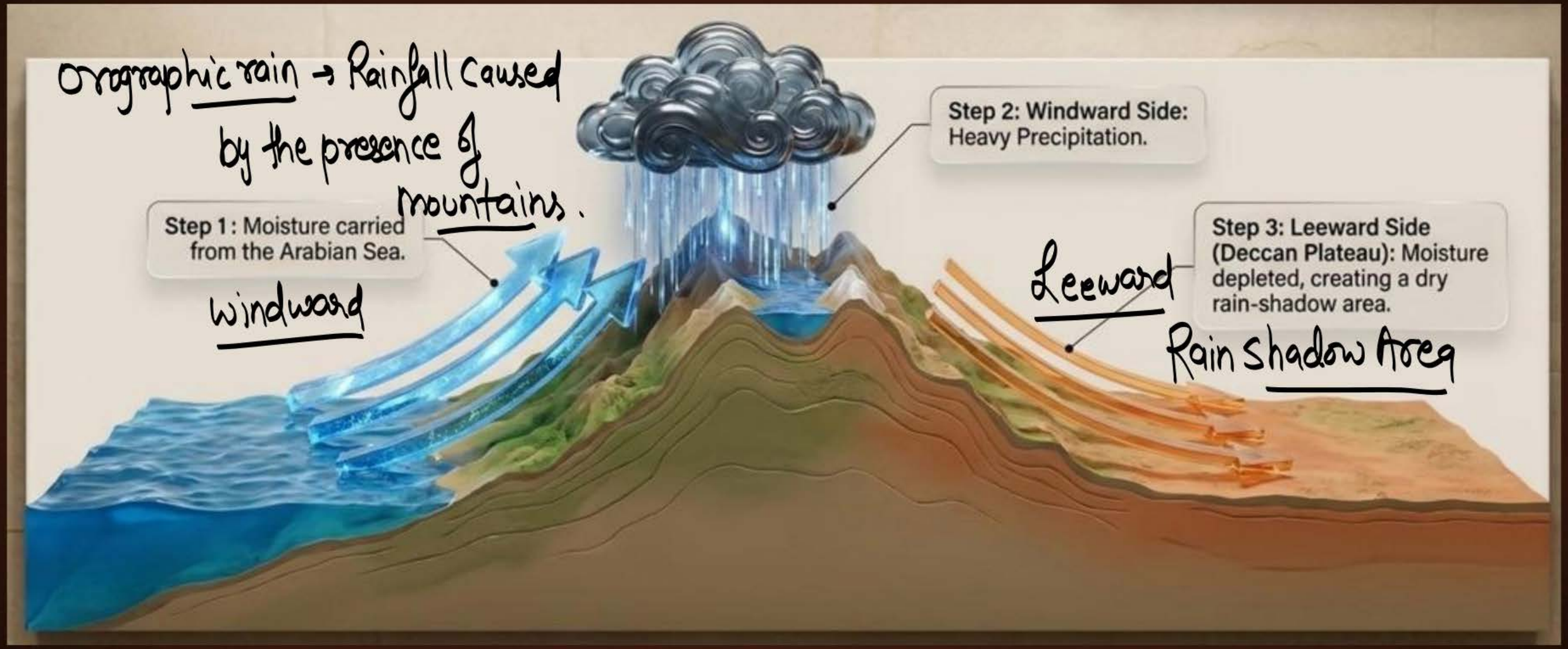
A Day

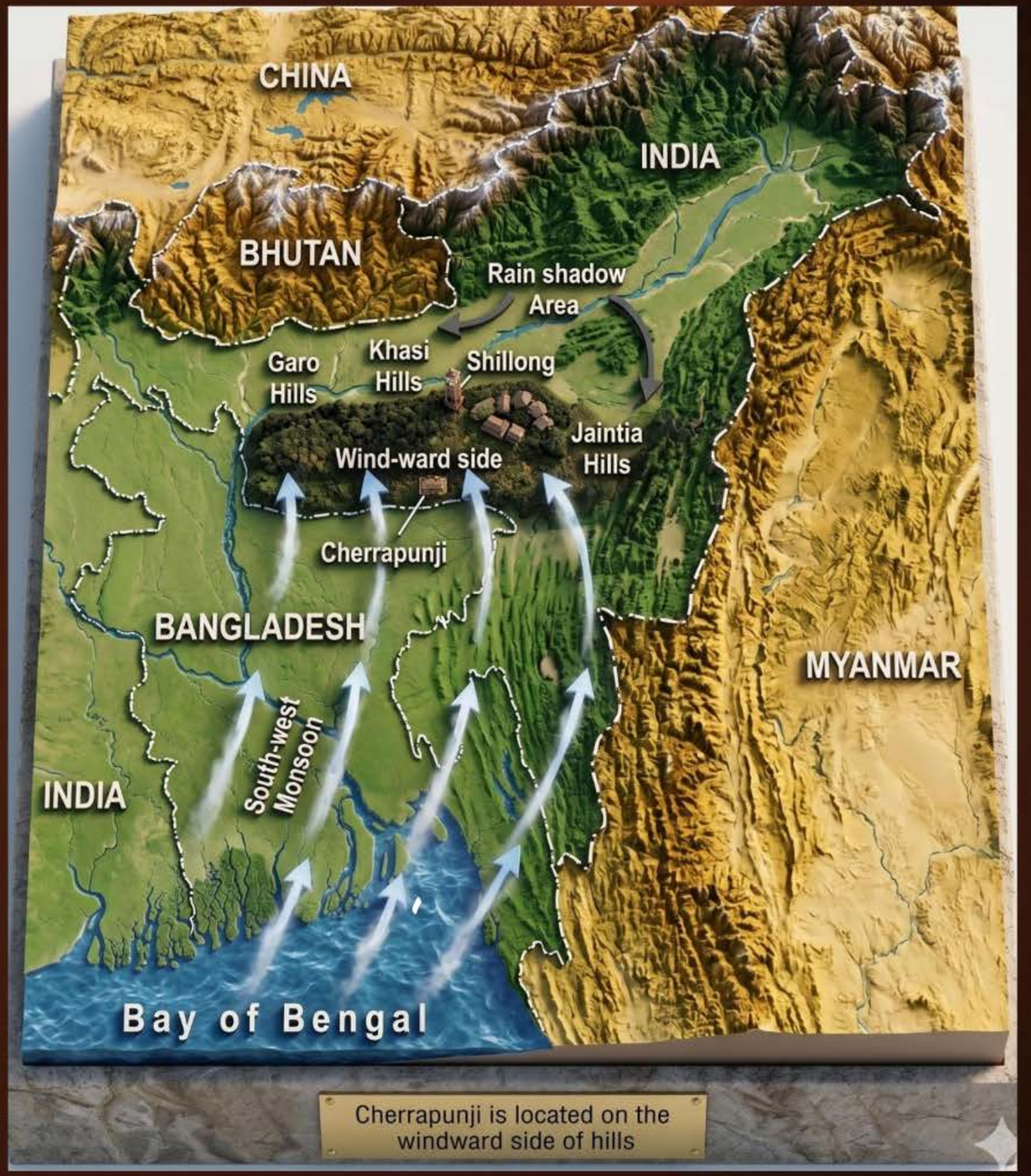
Bhopal — max - 42°C, min → 27°C → 15°C

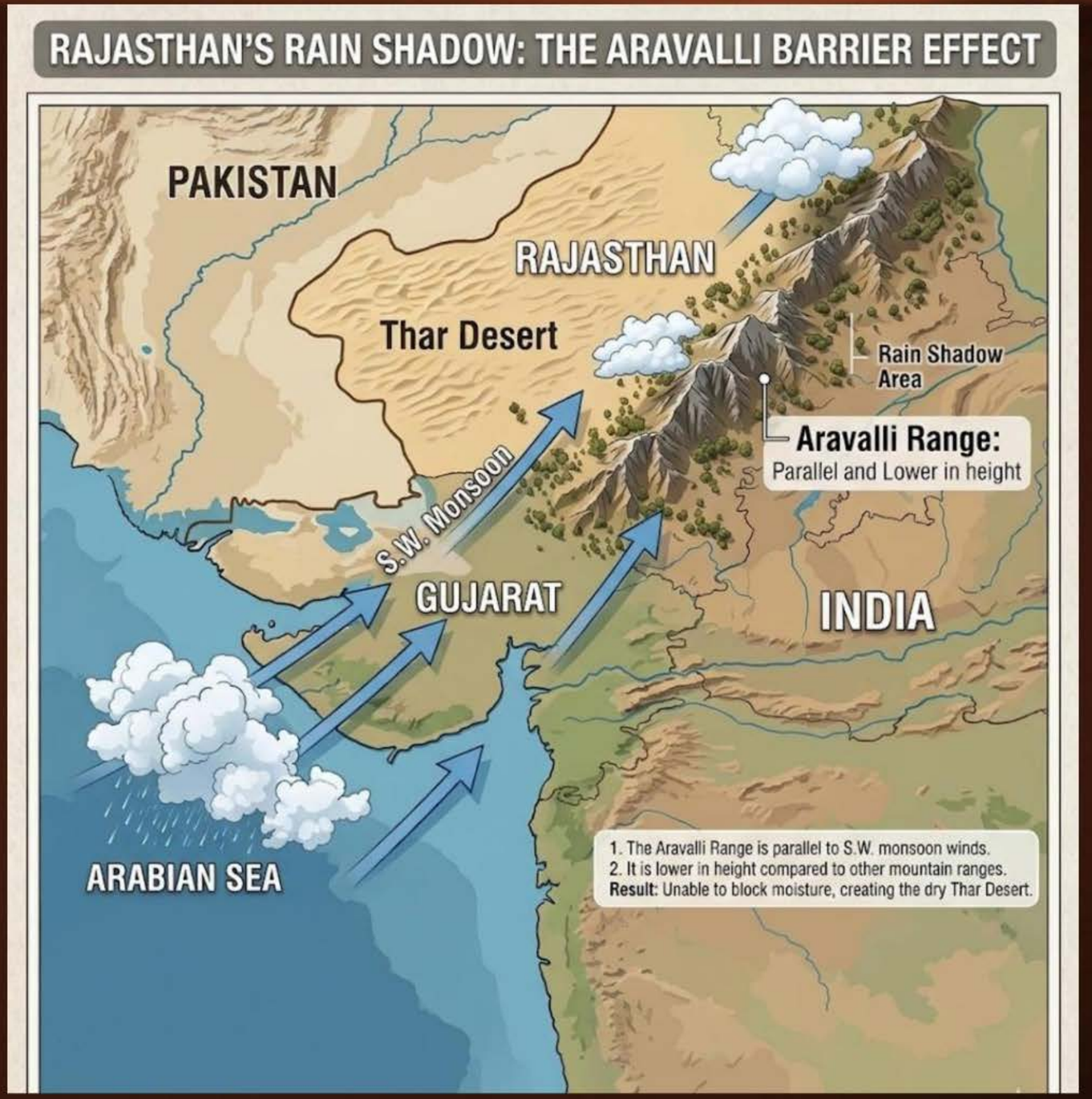
Range of yearly temperature → Difference between the Highest & lowest temperature of a place in a year.

Bhopal — max - 44°C, min → 11°C → 33°C


max - 50°C
min - 21°C
29°C
max - 31°C
min - 27°C
4°C


Orographic rain → Rainfall caused by the presence of mountains.
Step 1 : Moisture carried from the Arabian Sea.
windward
Step 2: Windward Side: Heavy Precipitation.
Leeward
Step 3: Leeward Side (Deccan Plateau): Moisture depleted, creating a dry rain-shadow area.
Rain Shadow Area


CHINA
INDIA
BHUTAN
Rain shadow Area
Garo Hills
Khasi Hills
Shillong
Jaintia Hills
Wind-ward side
Cherrapunji
BANGLADESH
MYANMAR
South-west Monsoon
INDIA
Bay of Bengal
Cherrapunji is located on the windward side of hills


RAJASTHAN'S RAIN SHADOW: THE ARAVALLI BARRIER EFFECT
PAKISTAN
RAJASTHAN
Thar Desert
Rain Shadow Area
Aravalli Range:
Parallel and Lower in height
S.W. Monsoon
GUJARAT
INDIA
ARABIAN SEA
1. The Aravalli Range is parallel to S.W. monsoon winds.
2. It is lower in height compared to other mountain ranges.
Result: Unable to block moisture, creating the dry Thar Desert.


INDIA
ARABIAN SEA
Windward side
Western Ghats
Rain Shadow Area
Eastern Ghats
Windward side
Chennai
N.E. Monsoon Winds
Mangalore
Chennai
BAY OF BENGAL
S.W. Monsoon Winds
INDIAN OCEAN

Factors Affecting the climate of India.

i) **Latitude** → Place nearby equator recieves maximum Sunlight and are very hot. As we move away from equator temperature decrease.

ii) **Altitude** → Temperature increases with height.

166m → 1°c less

166m - 1°c

iii) **Relief & topography** → Geographical location & land form.

iv) **Presence of Himalayan Mountains**
- → Stops cold siberian winds.
- → Stops South west monsoon winds in India.

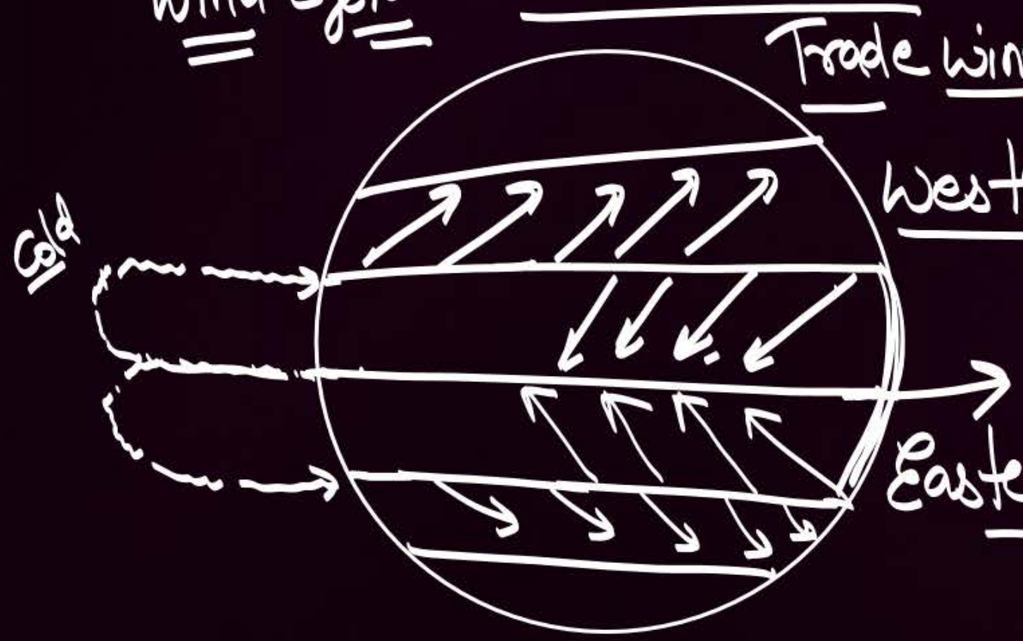
v) **Influence of the Sea**

vi) **Distance from the sea**

vii) **Monsoon winds.**

- Onset → South west winds
- Retreating → North East winds.

Wind System → Permanent winds

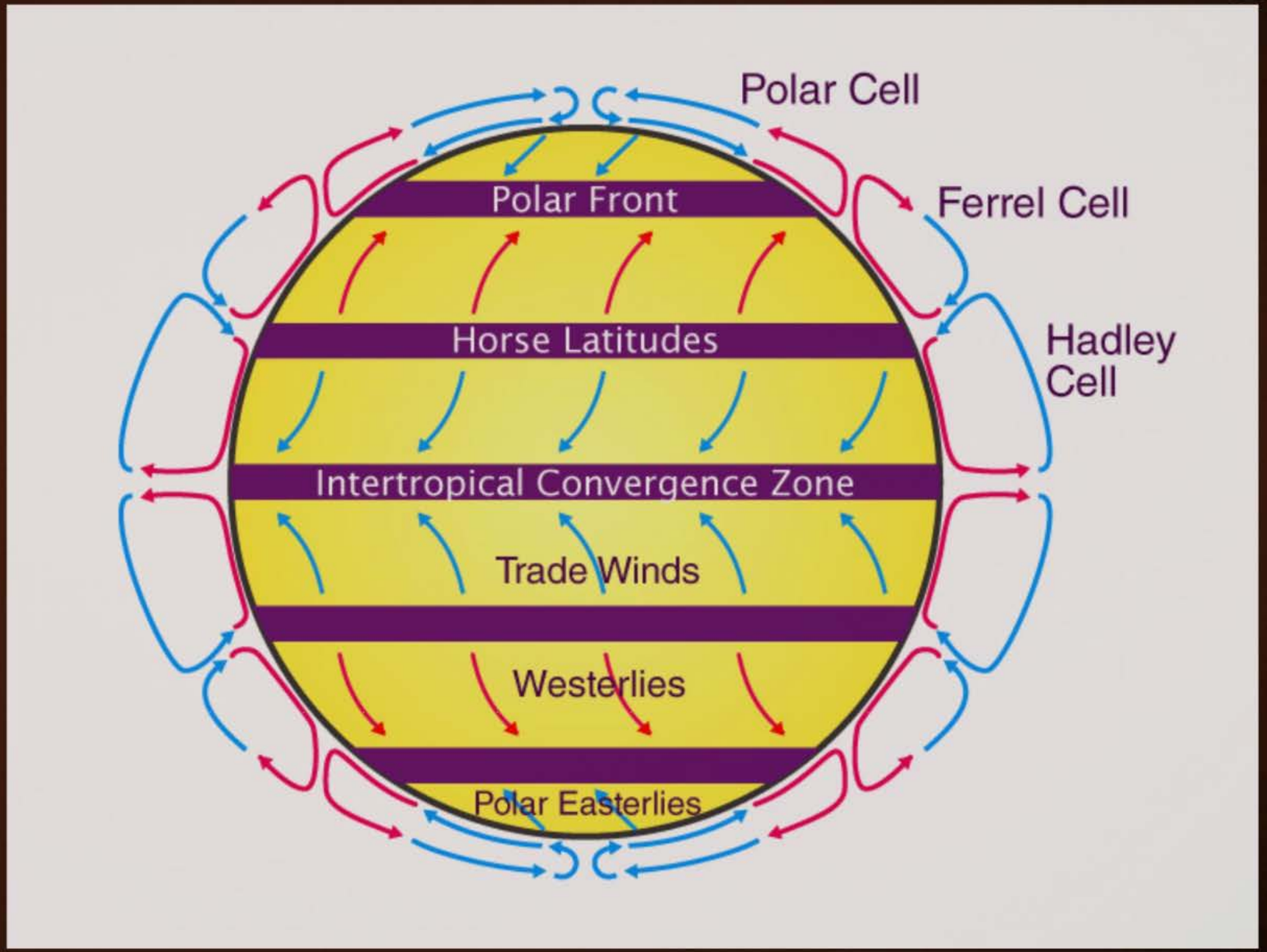


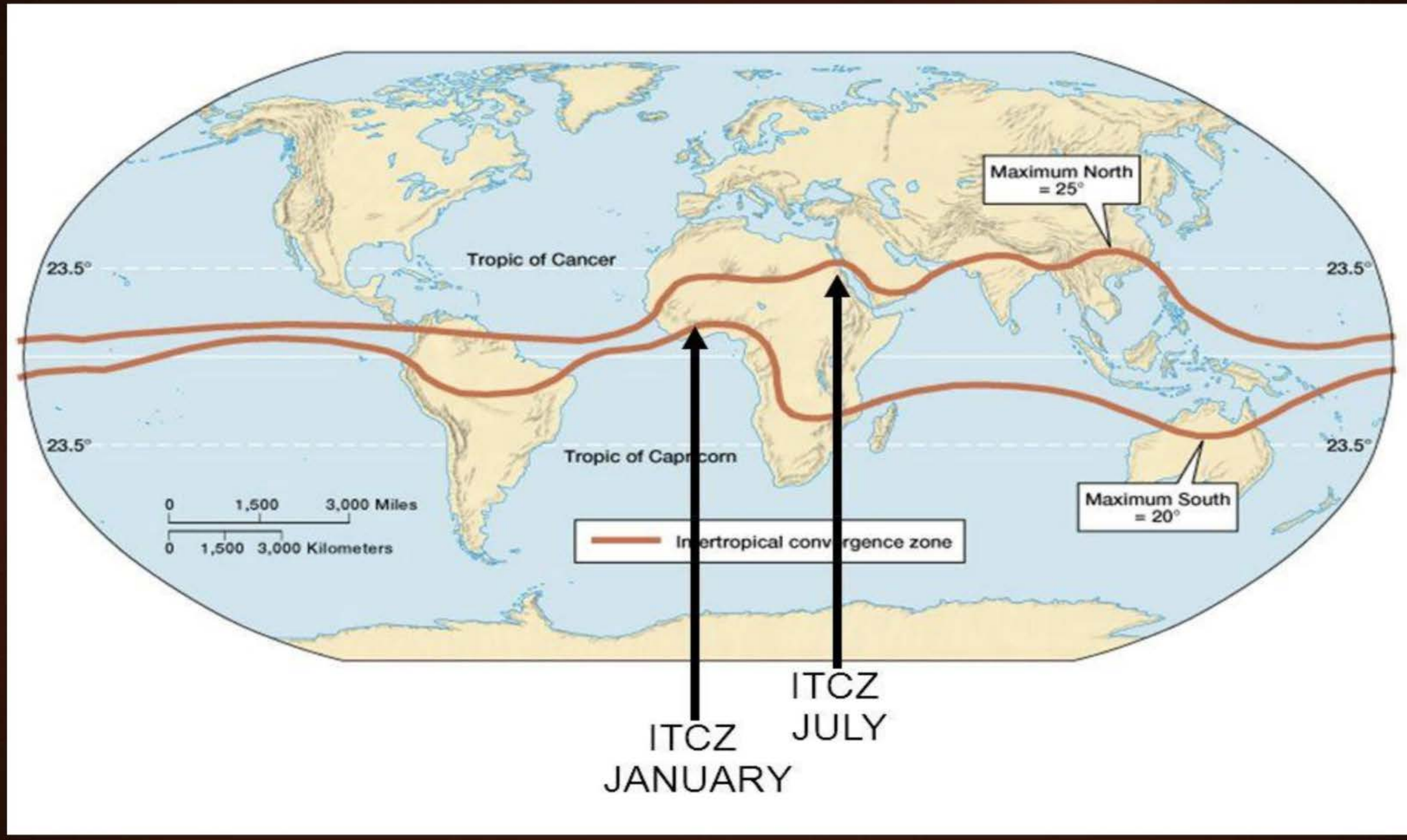
ITCZ → Intertropical Convergence zone

viii) Western Disturbance

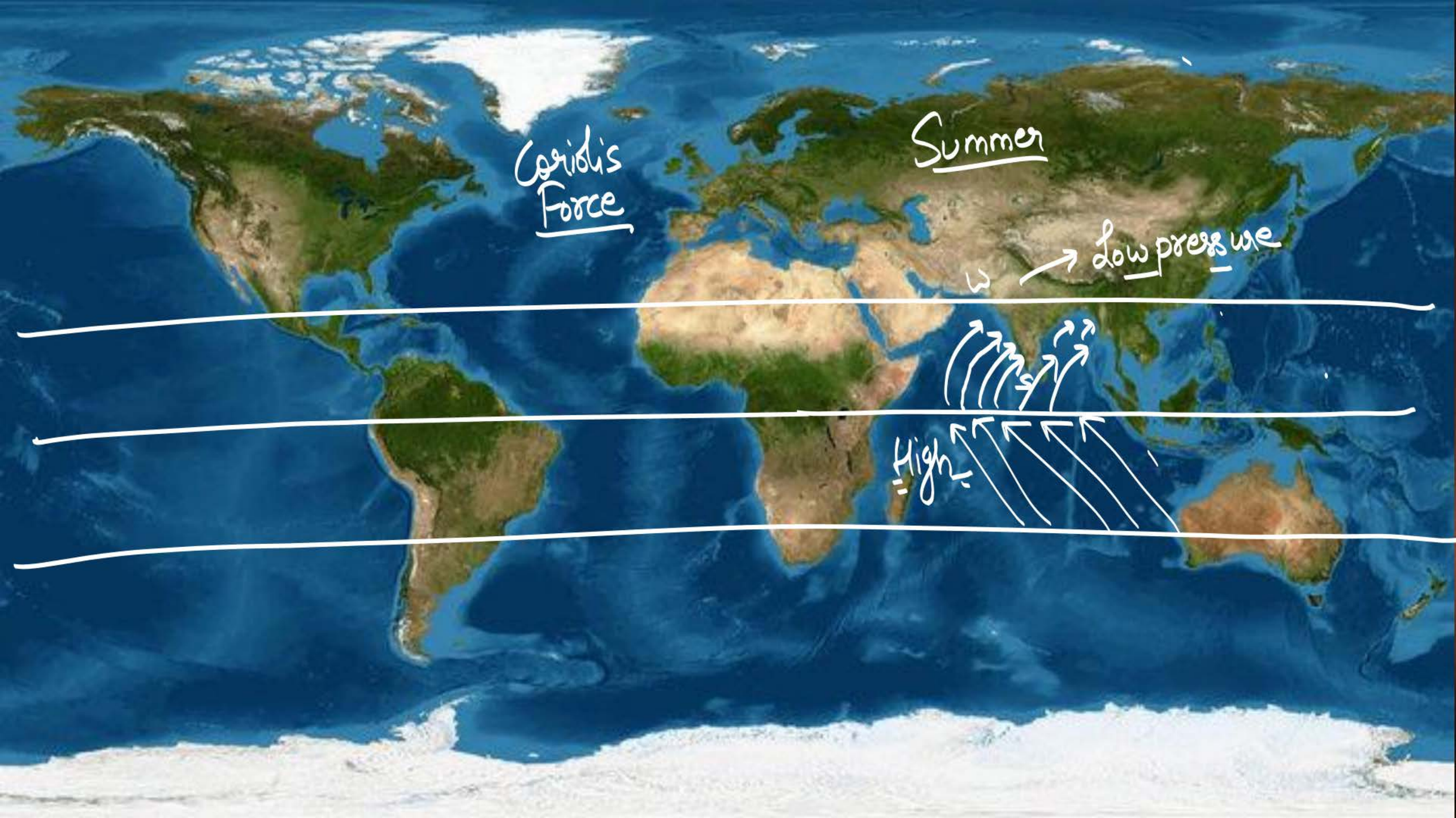
ix) Jet streams

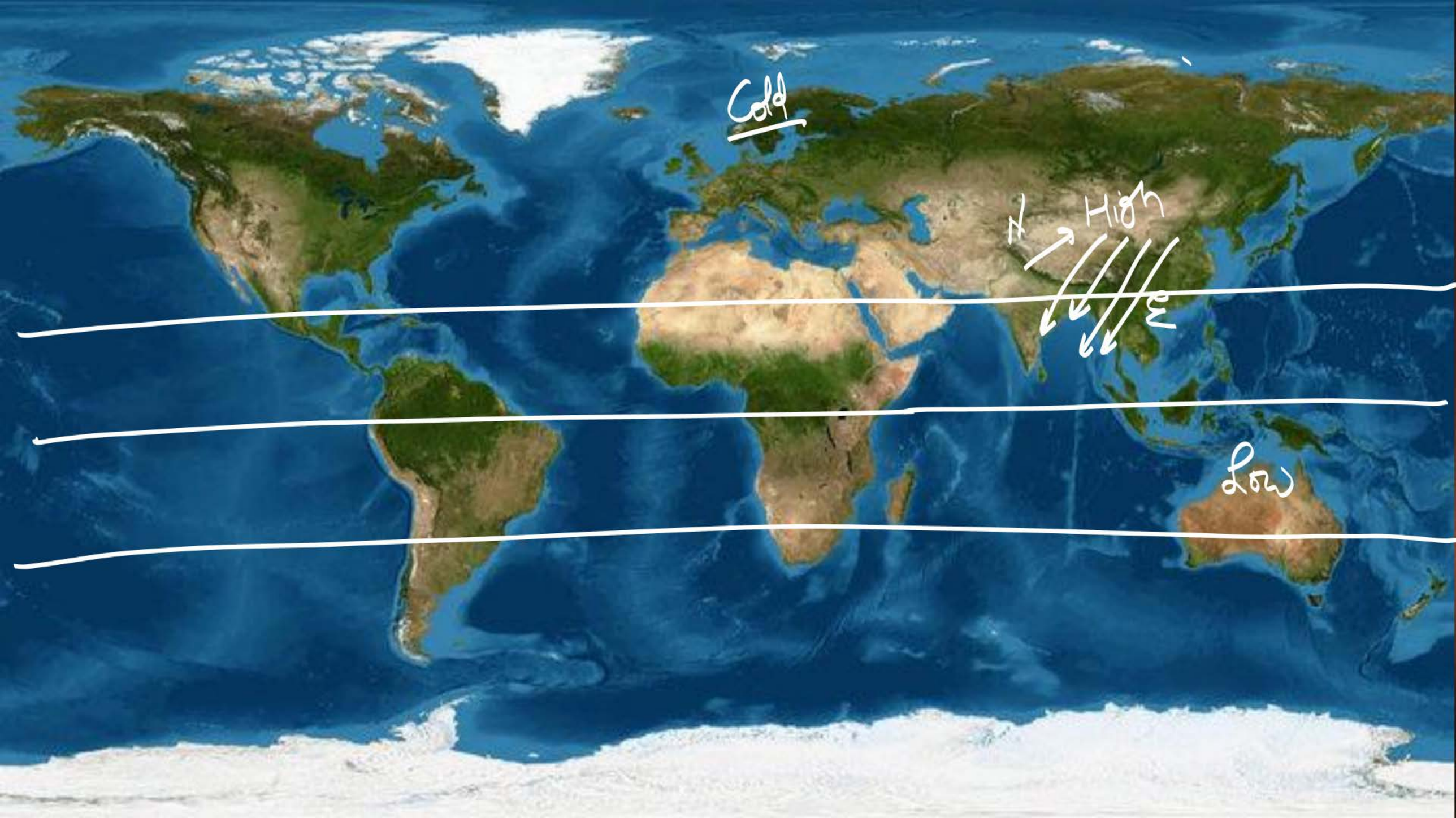
x) El - Nino

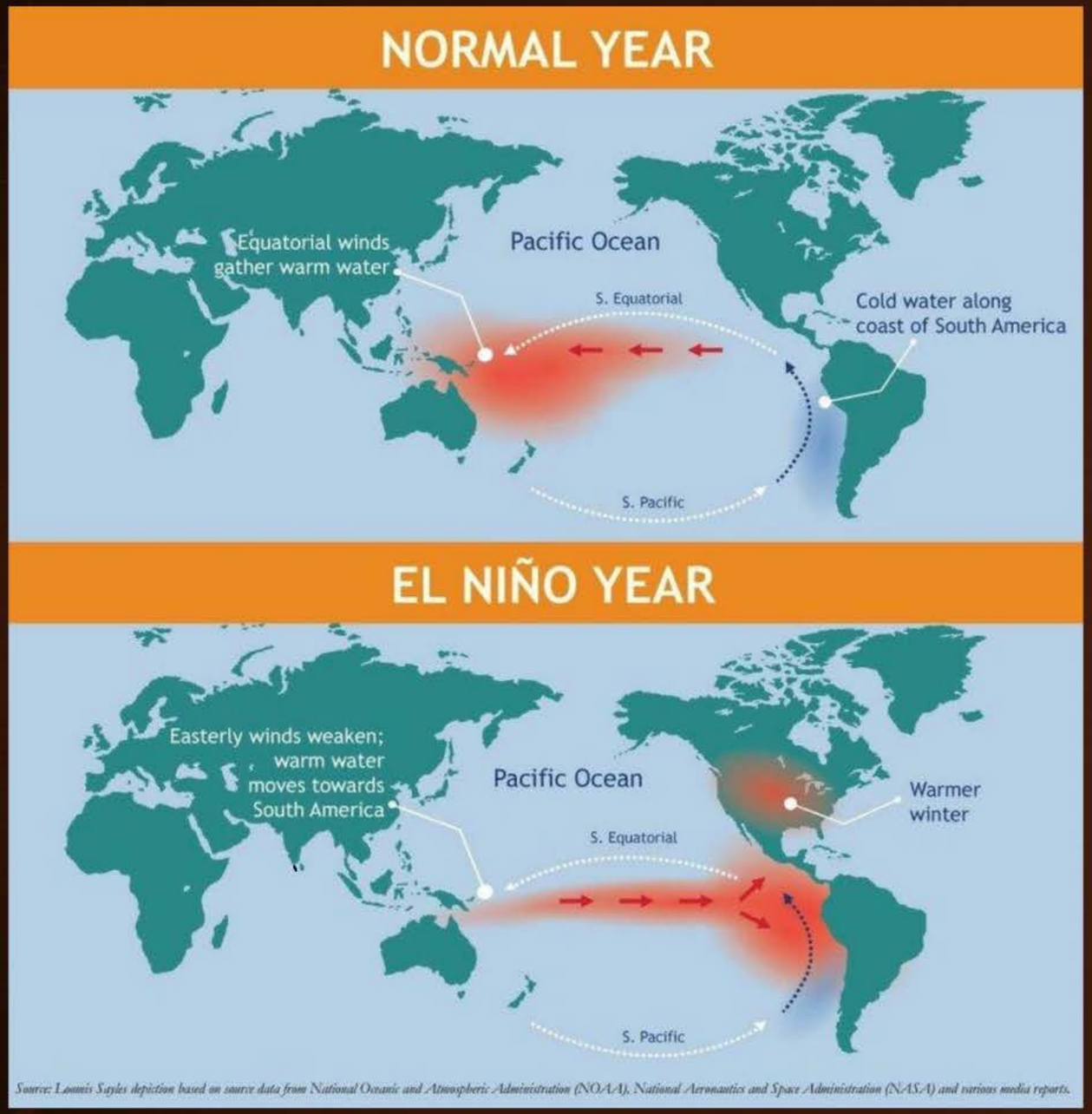

Polar Cell
Ferrel Cell
Hadley Cell
Polar Front
Horse Latitudes
Intertropical Convergence Zone
Trade Winds
Westerlies
Polar Easterlies

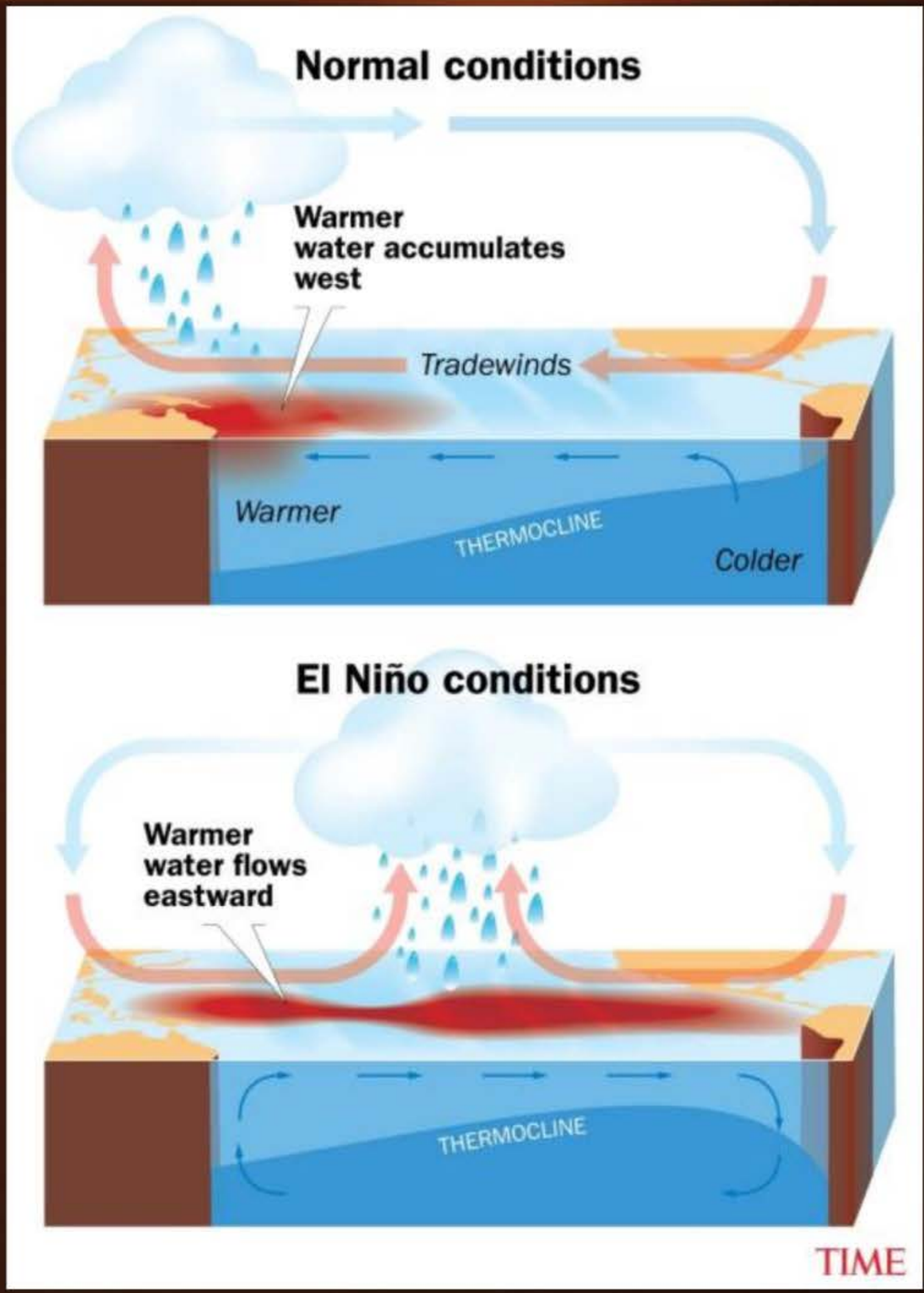

Maximum North
= 25°
Tropic of Cancer
23.5°
23.5°
23.5°
23.5°
Tropic of Capricorn
Maximum South
= 20°
0 1,500 3,000 Miles
0 1,500 3,000 Kilometers
Intertropical convergence zone
ITCZ
JANUARY
ITCZ
JULY


Jet Stream
Mediterranean
Western Disturbances
ARCTIC OCEAN
Greenland
(Denmark)
Svalbard
(Norway)
United States
Canada
Russia
Kazakhstan
Mongolia
China
India
Iran
Japan
N.Korea
S.Korea
Taiwan
Philippines
Malaysia
Sri Lanka
Nepal
Bhutan
Vietnam
Laos
Cambodia
Thailand
Bangl.
Myanmar
Pakistan
Afghan.
Uzbek.
Turkm.
Kyrgyzstan
Tajikistan
Armenia
Turkey
Cyprus
Syr.
Iraq
Kuwait
Qatar
Jordan
Saudi Arabia
U.A.E.
Oman
Yemen
Israel
Lebanon
Egypt
Libya
Algeria
Western Sahara
Morocco
Tunisia
Spain
France
Portugal
U.K.
Ireland
Iceland
Norway
Sweden
Finland
Estonia
Latvia
Lithuania
Belar.
Ukraine
Poland
Germany
Neth.
Belg.
Den.
Mexico
Caribbean
Central America
Guyana
Suriname
French Guiana
(France)
Venez.
Colombia
Ecuador
Peru
Brazil
Bolivia
Paraguay
Chile
Argentina
Uruguay
Senegal
Maurit.
Mali
Niger
Chad
Eritrea
Sudan
Djibouti
Ethiopia
South Sudan
The Gambia
Guinea-Bissau
Guinea
Sierra Leone
Liberia
Cote d'Ivoire
B.Faso
Ghana
Togo
Nigeria
Cam.
C.A.R.
Equatorial Guinea
Gabon
Congo
Dem. Rep. of Congo
Uganda
Rwanda
Burundi
Kenya
Somalia
Tanz.
Angola
Zambia
Zim.
Namibia
Bots.
South Africa
Lesotho
Malawi
Mozambique
eSwatini
Madagascar
Indonesia
Papua New Guinea
Solomon Islands
Timor-Leste
Australia
New Caledonia
Fiji
New Zealand
PACIFIC OCEAN
ATLANTIC OCEAN
INDIAN OCEAN
SOUTHERN OCEAN
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Coriolis Force
Summer
Low pressure
High


Cold
N
High
Low


NORMAL YEAR
Equatorial winds gather warm water
Pacific Ocean
S. Equatorial
Cold water along coast of South America
S. Pacific
EL NIÑO YEAR
Easterly winds weaken; warm water moves towards South America
Pacific Ocean
Warmer winter
S. Equatorial
S. Pacific
Source: Loomis Sayles depiction based on source data from National Oceanic and Atmospheric Administration (NOAA), National Aeronautics and Space Administration (NASA) and various media reports.

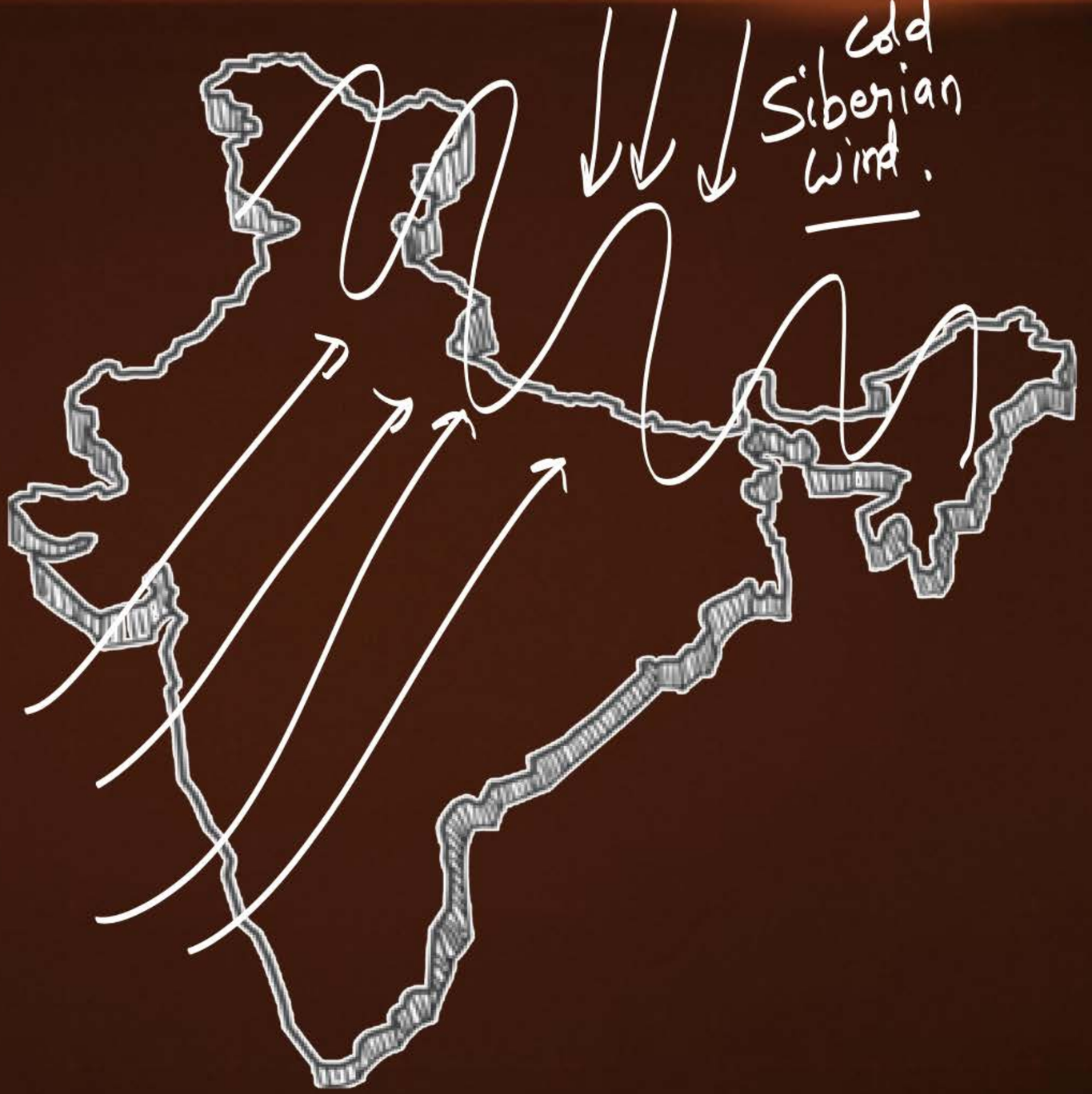

Normal conditions
Warmer water accumulates west
Tradewinds
Warmer
THERMOCLINE
Colder
El Niño conditions
Warmer water flows eastward
THERMOCLINE
TIME

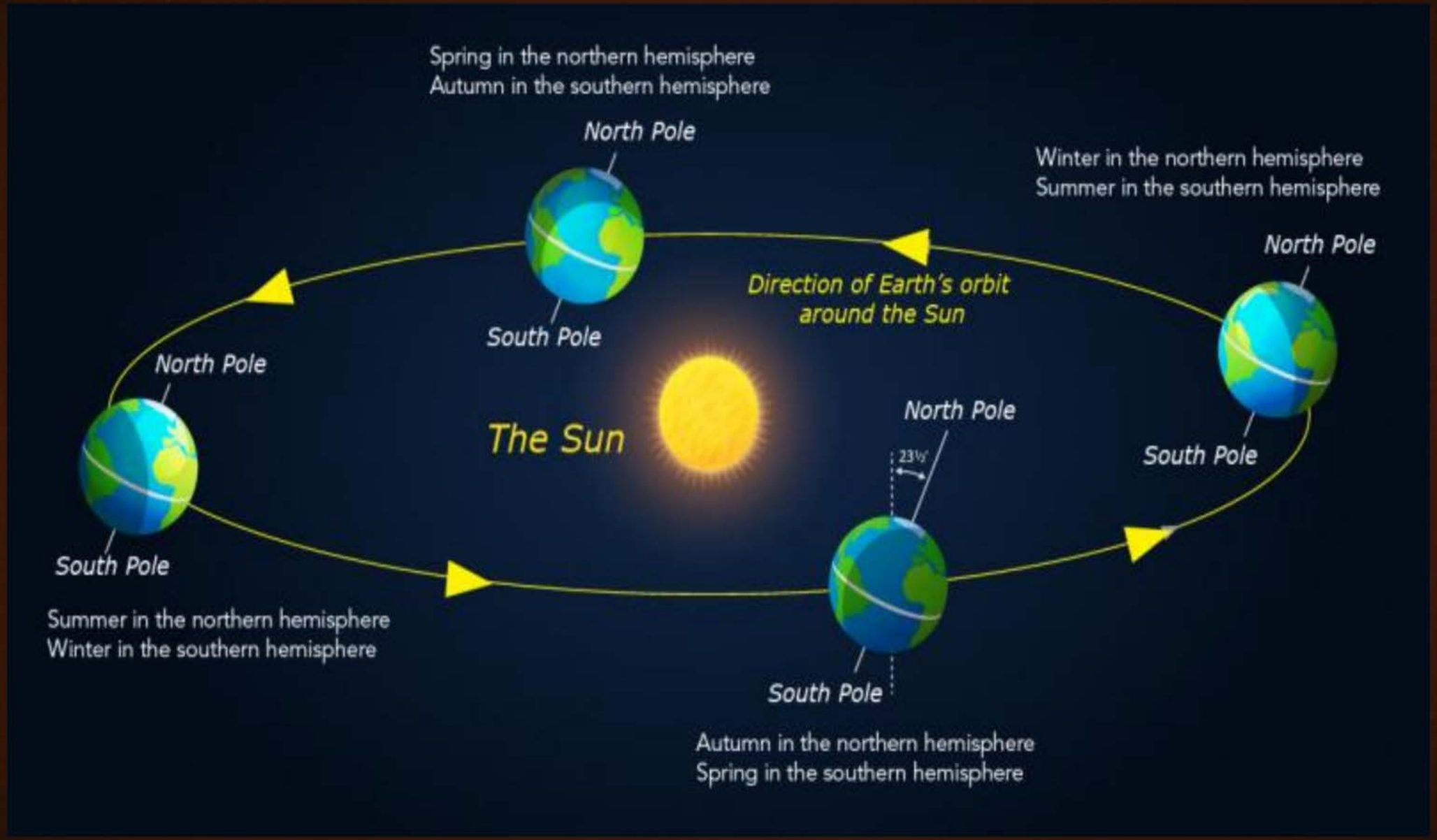
i) Summer → March - May

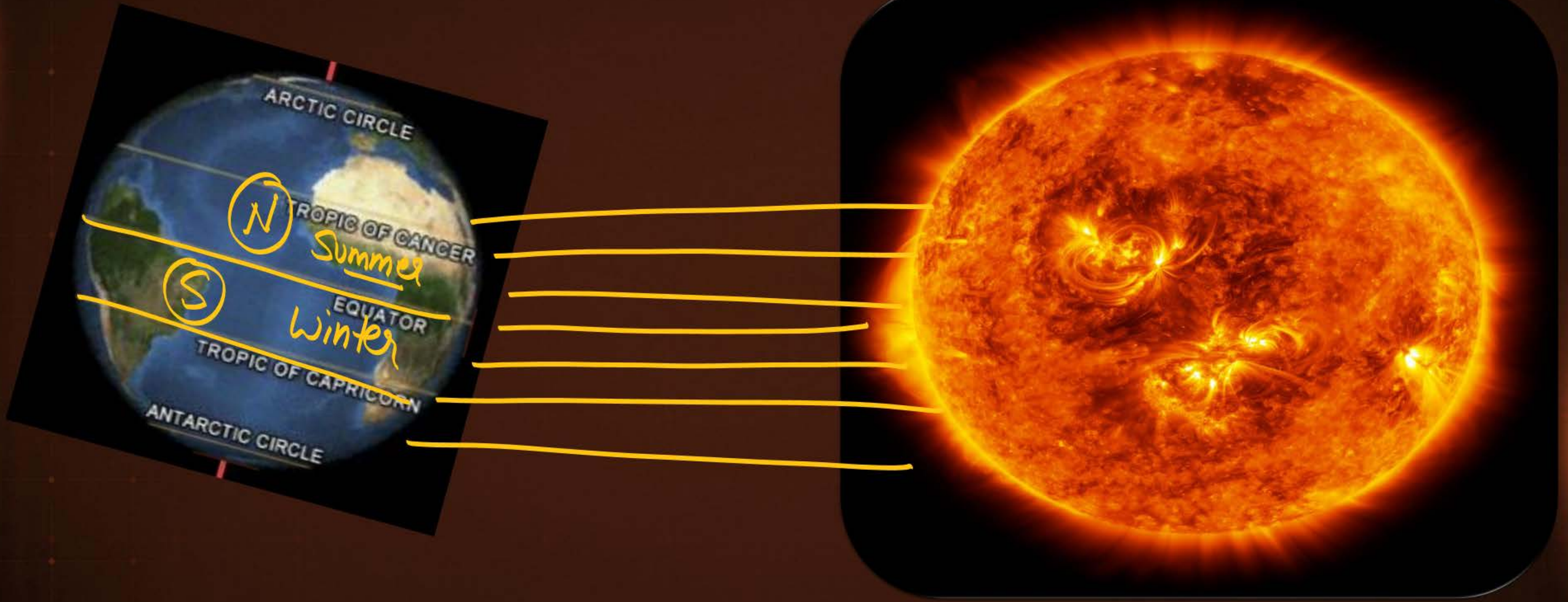
ii) Onset of Monsoon → June - Sept

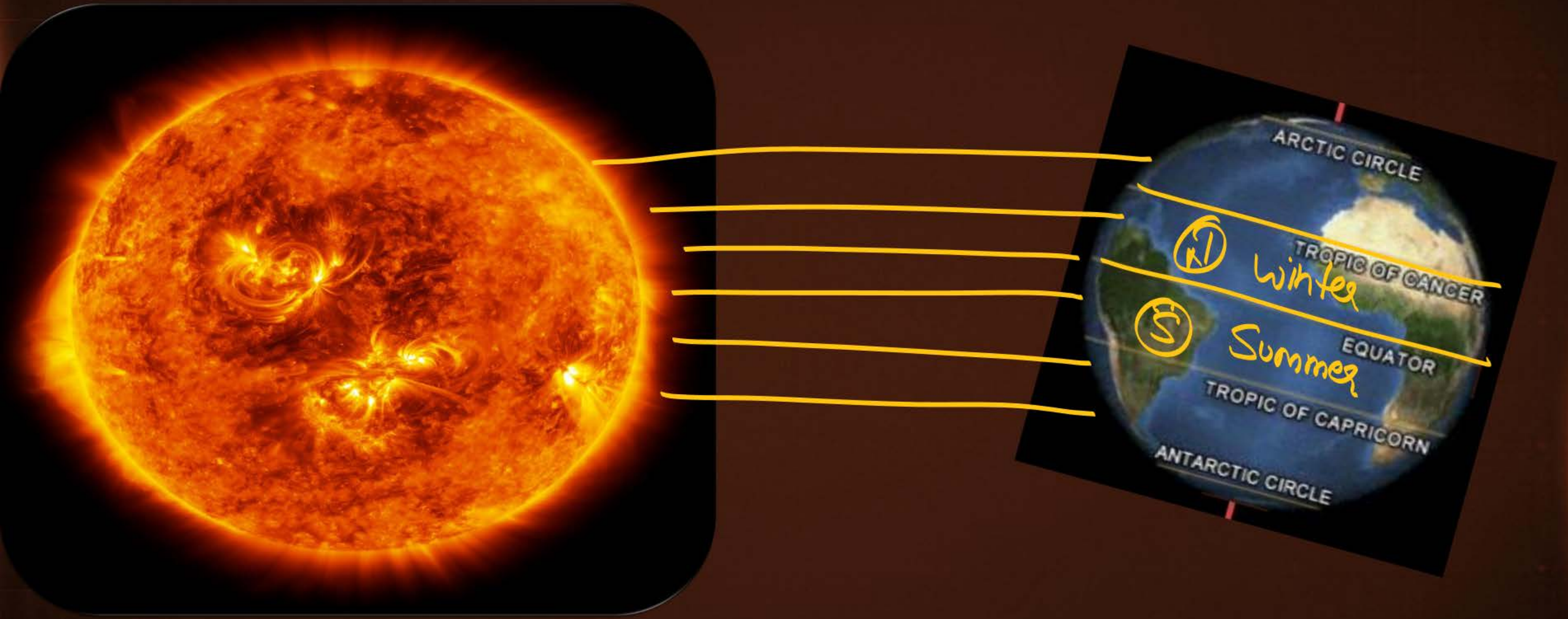
iii) Retreating Monsoon → Oct - Nov

iv) Cold → Dec - Feb → Western Disturbances

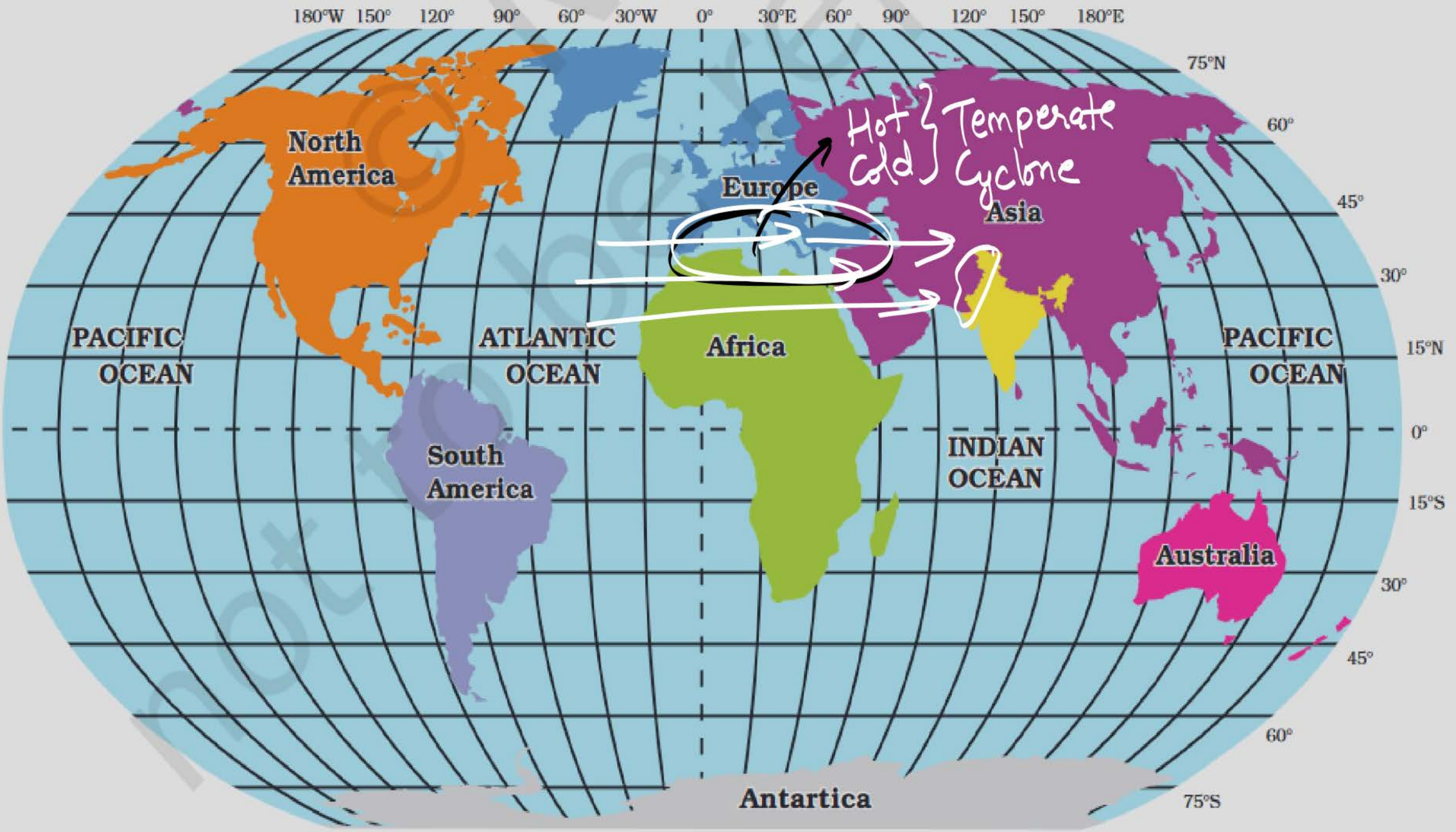

cold
Siberian
wind.

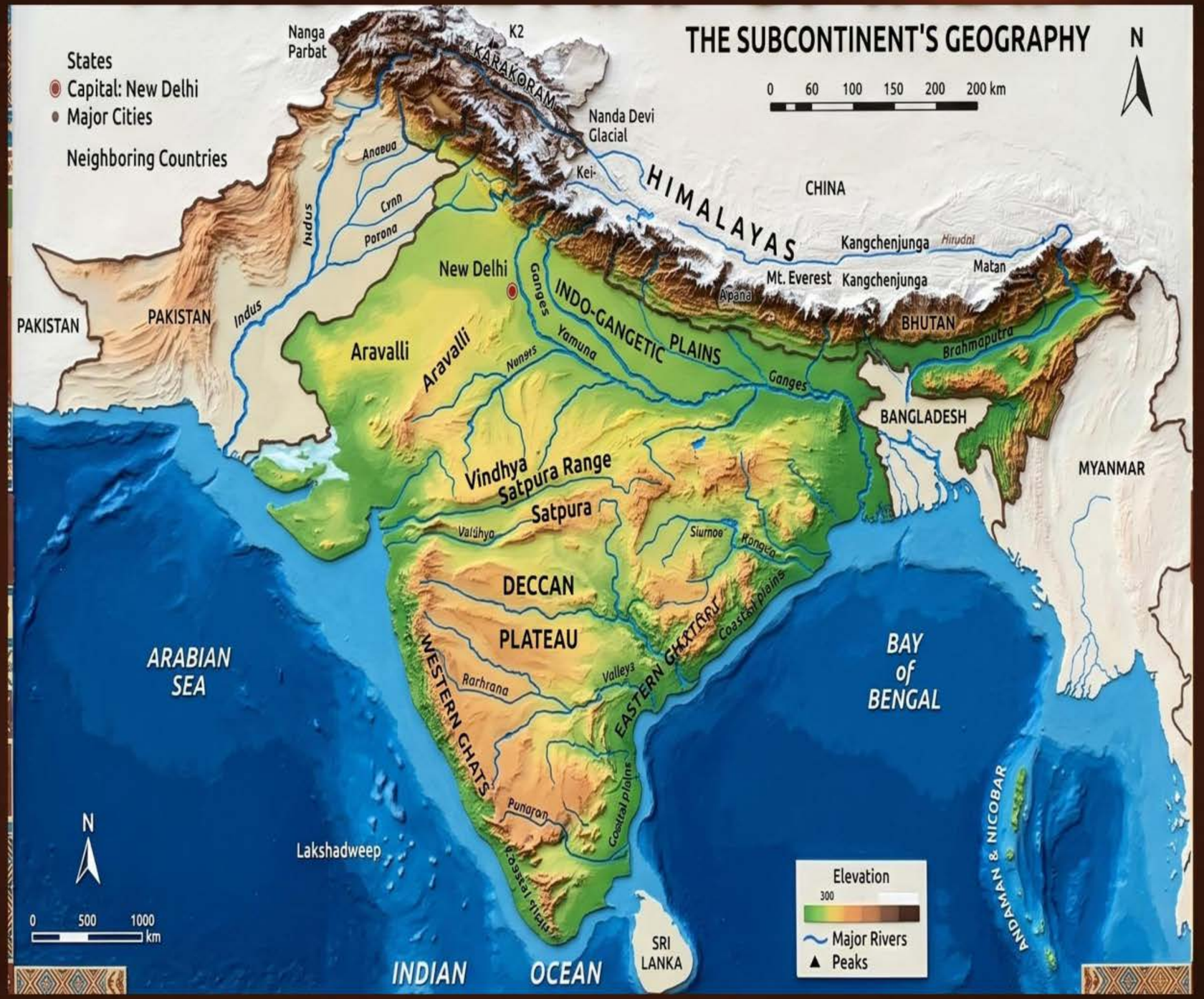

Spring in the northern hemisphere
Autumn in the southern hemisphere
North Pole
South Pole
Winter in the northern hemisphere
Summer in the southern hemisphere
North Pole
South Pole
Direction of Earth's orbit around the Sun
North Pole
South Pole
The Sun
North Pole
23½°
South Pole
Summer in the northern hemisphere
Winter in the southern hemisphere
Autumn in the northern hemisphere
Spring in the southern hemisphere


ARCTIC CIRCLE
N
TROPIC OF CANCER
Summer
EQUATOR
S
Winter
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE


ARCTIC CIRCLE
TROPIC OF CANCER
Winter
EQUATOR
Summer
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE

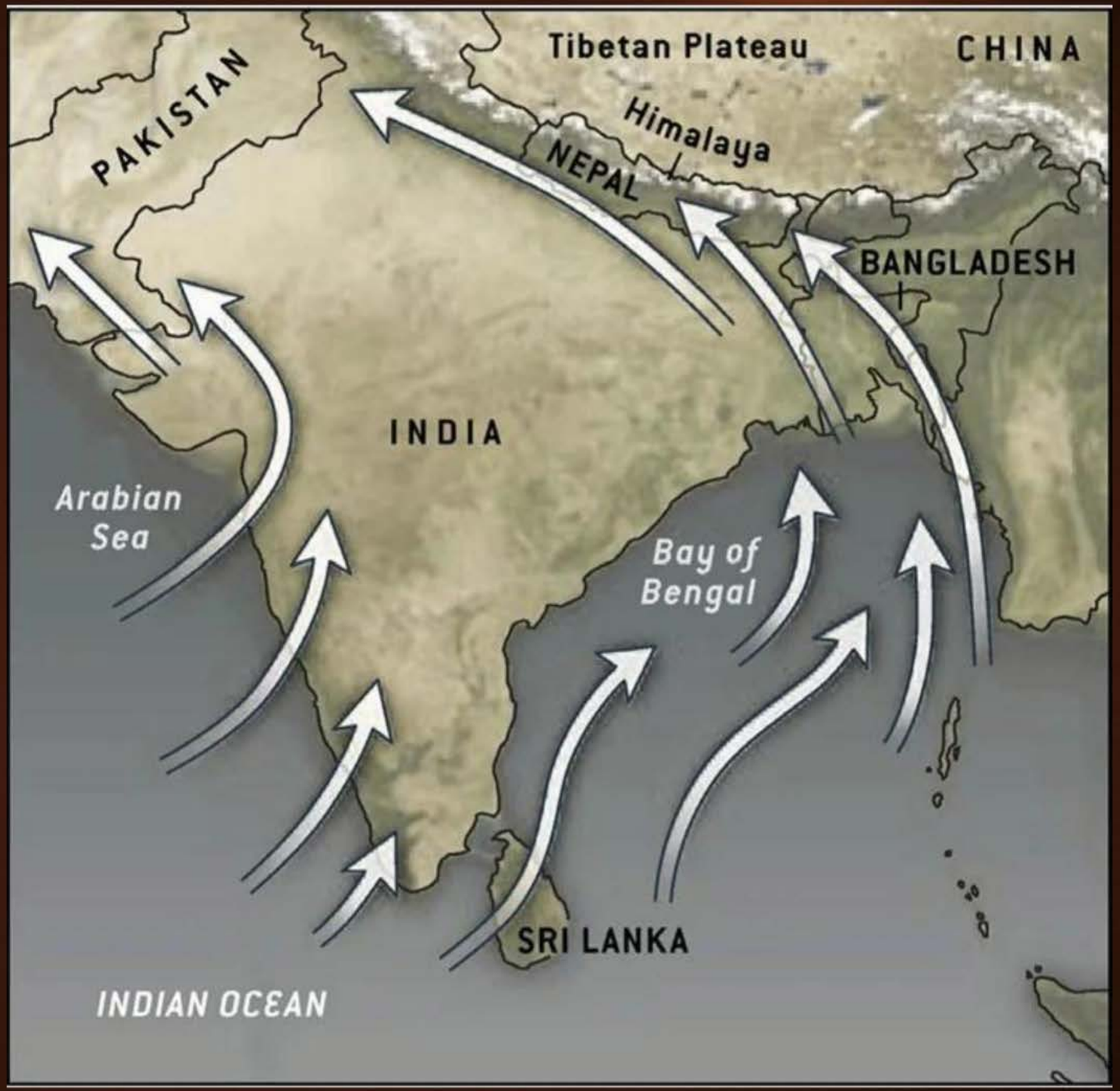

INDIA
States and Union Territories
INDIA FLAG
TAJIKISTAN
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR (BURMA)
SRI LANKA
LADAKH
Srinagar
Leh
JAMMU & KASHMIR
Jammu
HIMACHAL PRADESH
Shimla
PUNJAB
CHANDIGARH
Dehradun
UTTARAKHAND
HARYANA
DELHI
New Delhi
Jaipur
RAJASTHAN
UTTAR PRADESH
Lucknow
SIKKIM
Gangtok
ARUNACHAL PRADESH
Itanagar
Dispur
ASSAM
NAGALAND
Kohima
Shillong
MEGHALAYA
Imphal
MANIPUR
Agartala
TRIPURA
Aizawl
MIZORAM
BIHAR
Patna
JHARKHAND
Ranchi
WEST BENGAL
Kolkata
Gandhinagar
GUJARAT
Bhopal
MADHYA PRADESH
CHHATTISGARH
Raipur
ODISHA
Bhubaneswar
Daman
DADRA & NAGAR HAVELI AND DAMAN & DIU
MAHARASHTRA
Mumbai
TELANGANA
Hyderabad
Amaravati
Yanam (Puducherry)
ANDHRA PRADESH
Panaji
GOA
KARNATAKA
Bengaluru
Chennai
Mahe (Puducherry)
PUDUCHERRY
Kavaratti
LAKSHADWEEP (INDIA)
KERALA
TAMIL NADU
Karaikal (Puducherry)
Thiruvananthapuram
Palk Strait
ANDAMAN & NICOBAR ISLANDS (INDIA)
Port Blair
BAY OF BENGAL
ARABIAN SEA
INDIAN OCEAN
International Boundary
State/UT Boundary
Country Capital
State/UT Capital
Map not to Scale
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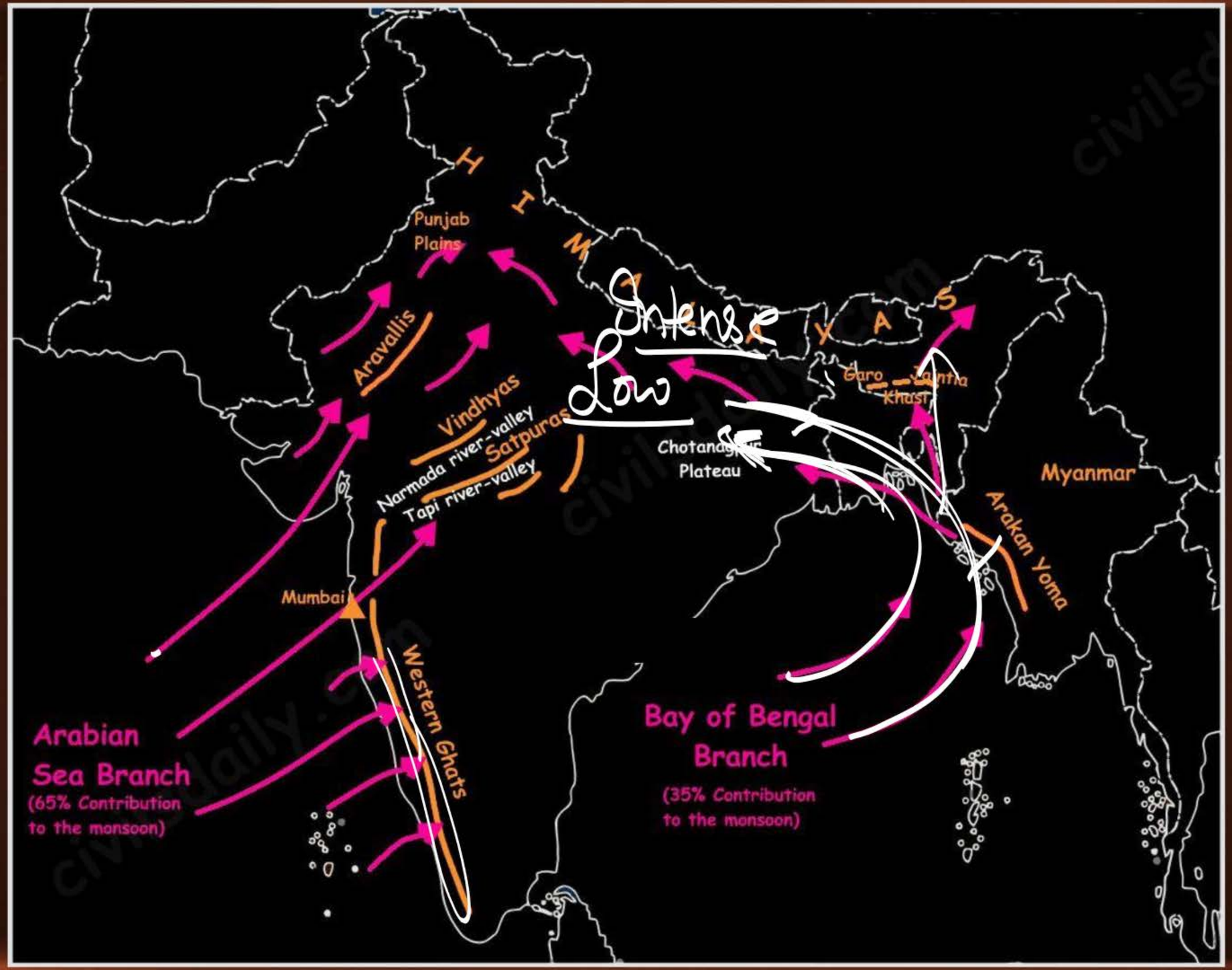

180°W 150° 120° 90° 60° 30°W 0° 30°E 60° 90° 120° 150° 180°E
75°N
60°
45°
30°
15°N
0°
15°S
30°
45°
60°
75°S
North America
Europe
Asia
Hot
cold
Temperate
Cyclone
PACIFIC OCEAN
ATLANTIC OCEAN
Africa
PACIFIC OCEAN
INDIAN OCEAN
South America
Australia
Antartica


THE SUBCONTINENT'S GEOGRAPHY
States
Capital: New Delhi
Major Cities
Neighboring Countries
Nanga Parbat
K2
KARAKORAM
Nanda Devi
Glacial
HIMALAYAS
CHINA
Kangchenjunga
Mt. Everest
Kangchenjunga
Matan
BHUTAN
Brahmaputra
New Delhi
Ganges
INDO-GANGETIC PLAINS
Yamuna
Ganges
PAKISTAN
PAKISTAN
Indus
Aravalli
Aravalli
BANGLADESH
MYANMAR
Vindhya
Satpura Range
Satpura
DECCAN
PLATEAU
WESTERN GHATS
EASTERN GHATS
Coastal plains
ARABIAN SEA
BAY of BENGAL
Lakshadweep
ANDAMAN & NICOBAR
SRI LANKA
INDIAN OCEAN
Elevation
Major Rivers
Peaks
0 60 100 150 200 200 km
0 500 1000 km
N

Arabian Sea

Bay of Bengal

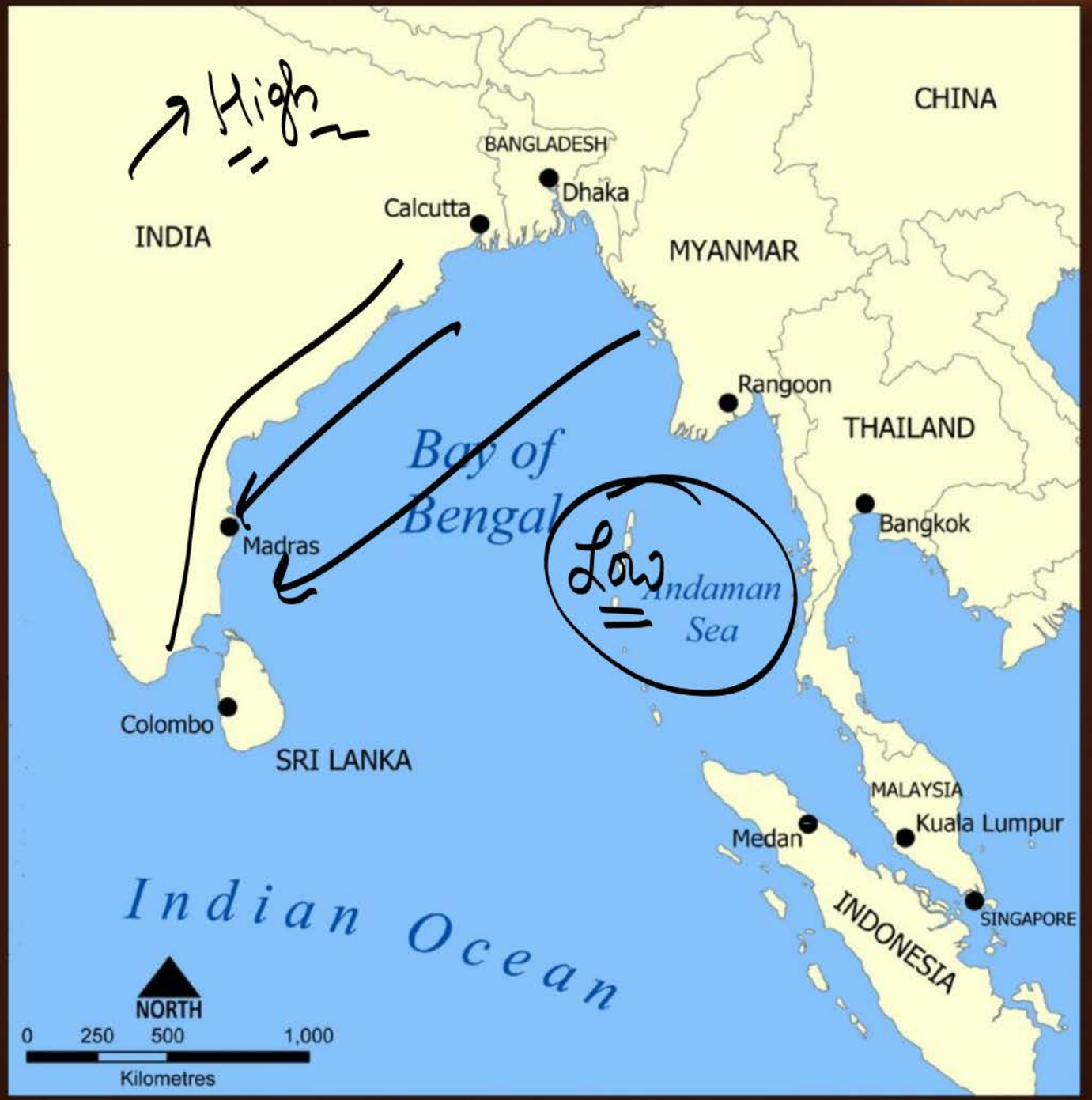


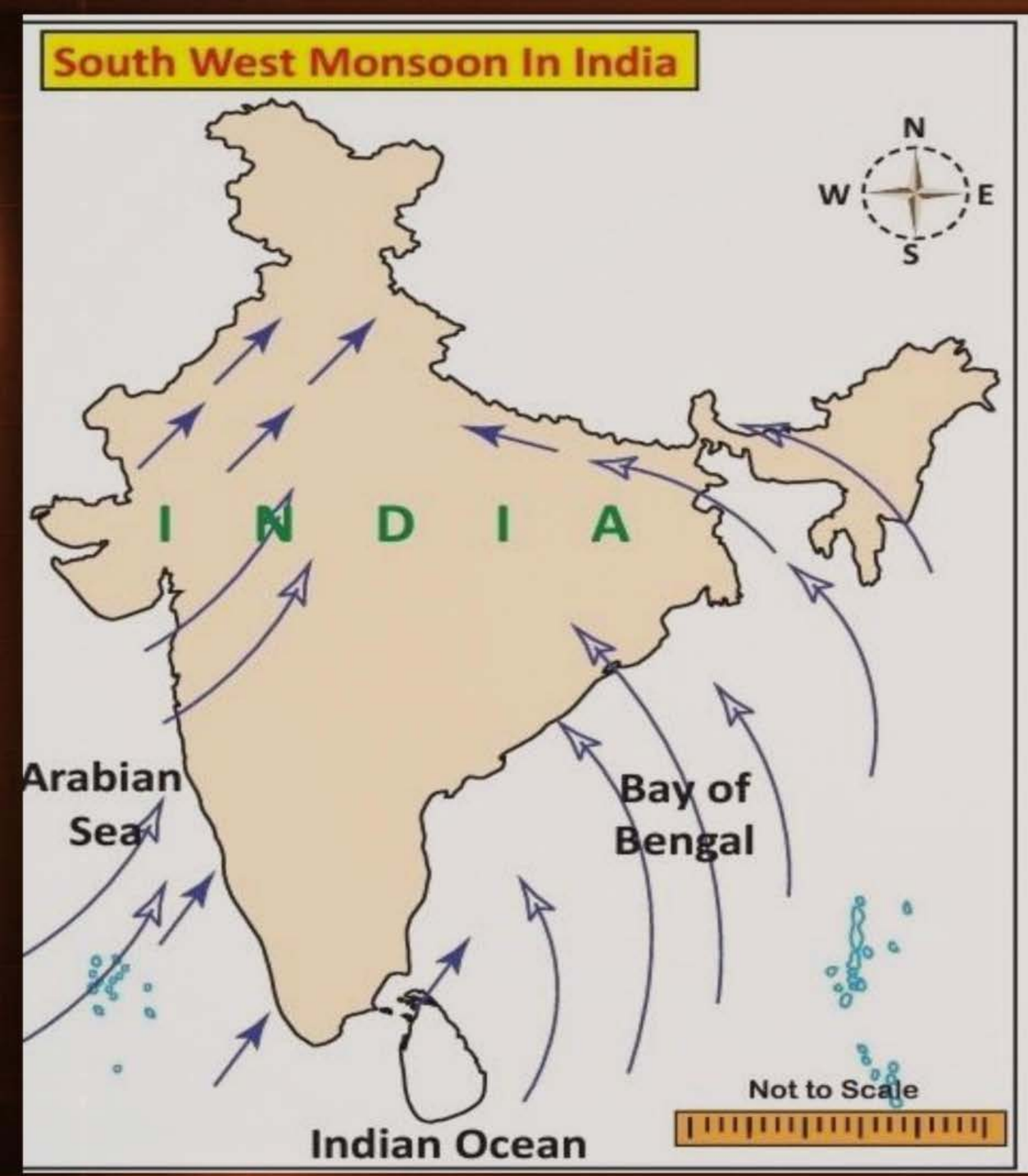

Punjab Plains
HIMALAYAS
Aravallis
Vindhyas
Narmada river-valley
Satpuras
Tapi river-valley
Intense Low
Chotanagpur Plateau
Garo
Khasi
Myanmar
Arakan Yoma
Mumbai
Western Ghats
Arabian Sea Branch
(65% Contribution to the monsoon)
Bay of Bengal Branch
(35% Contribution to the monsoon)



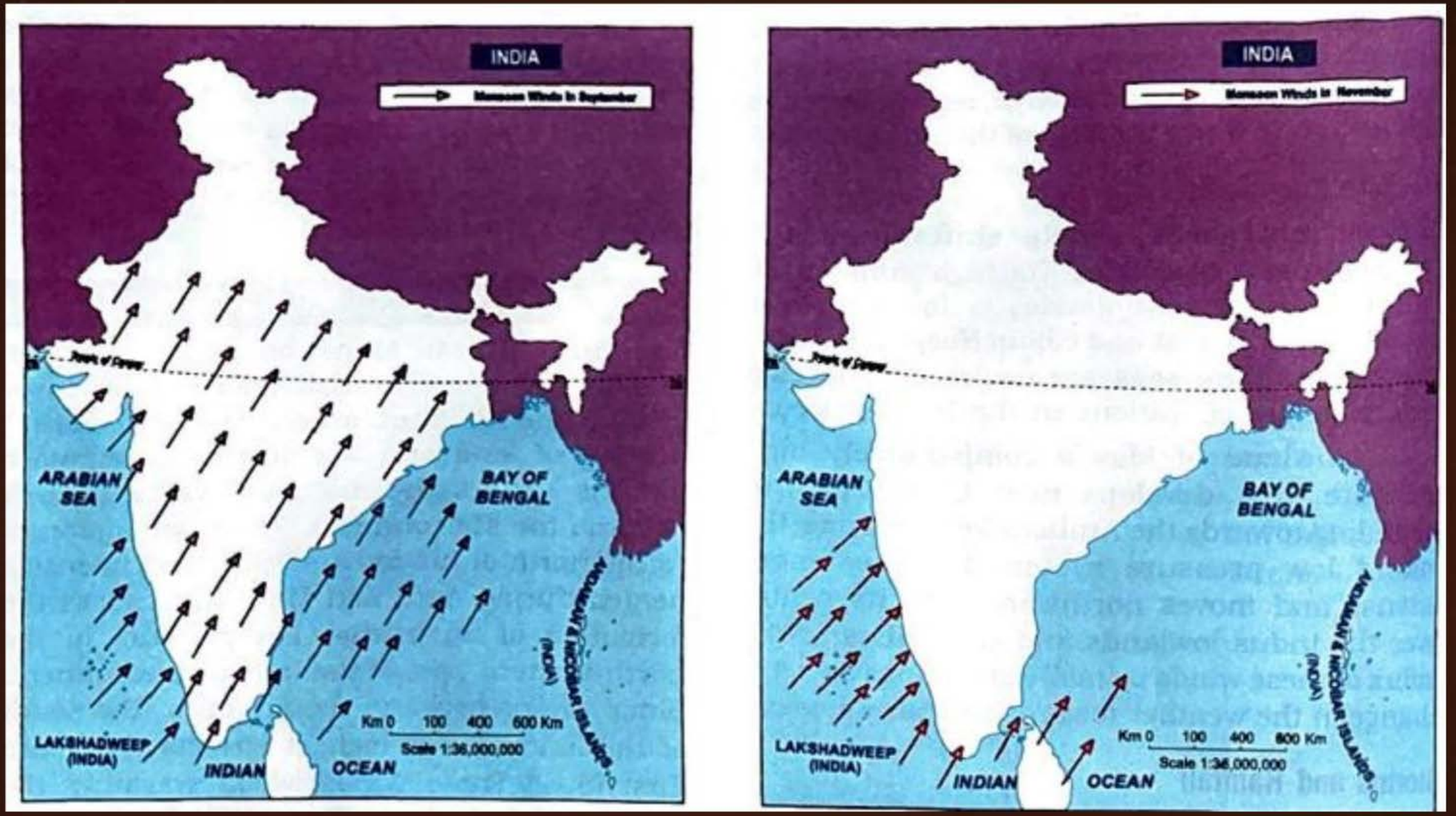
Tropical → Water

Temperate → Land

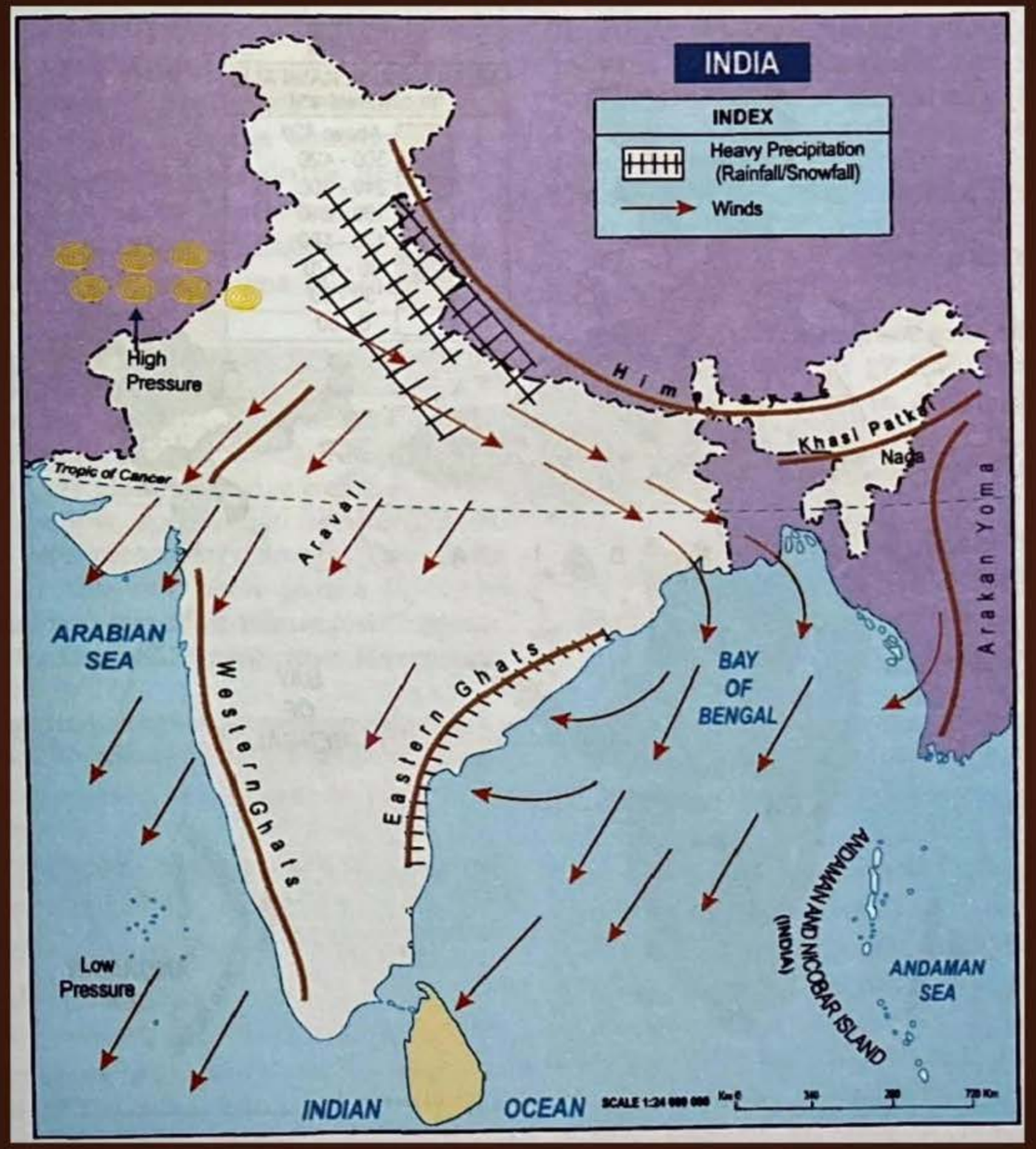



South West Monsoon In India
N
W
E
S
I N D I A
Arabian Sea
Bay of Bengal
Indian Ocean
Not to Scale

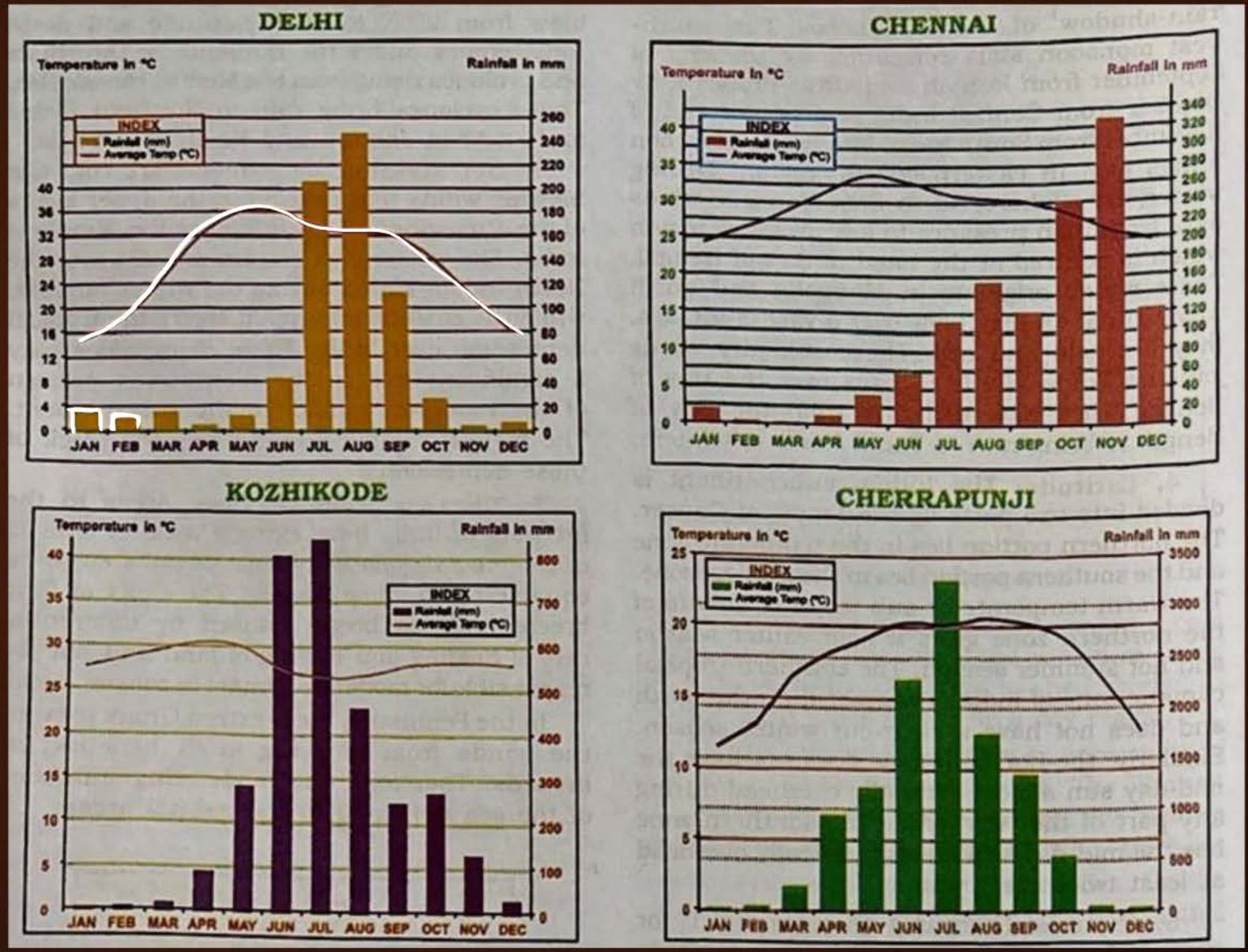

North East Monsoon In India
N
W
E
S
I N D I A
Arabian Sea
Bay of Bengal
Indian Ocean
Not to Scale

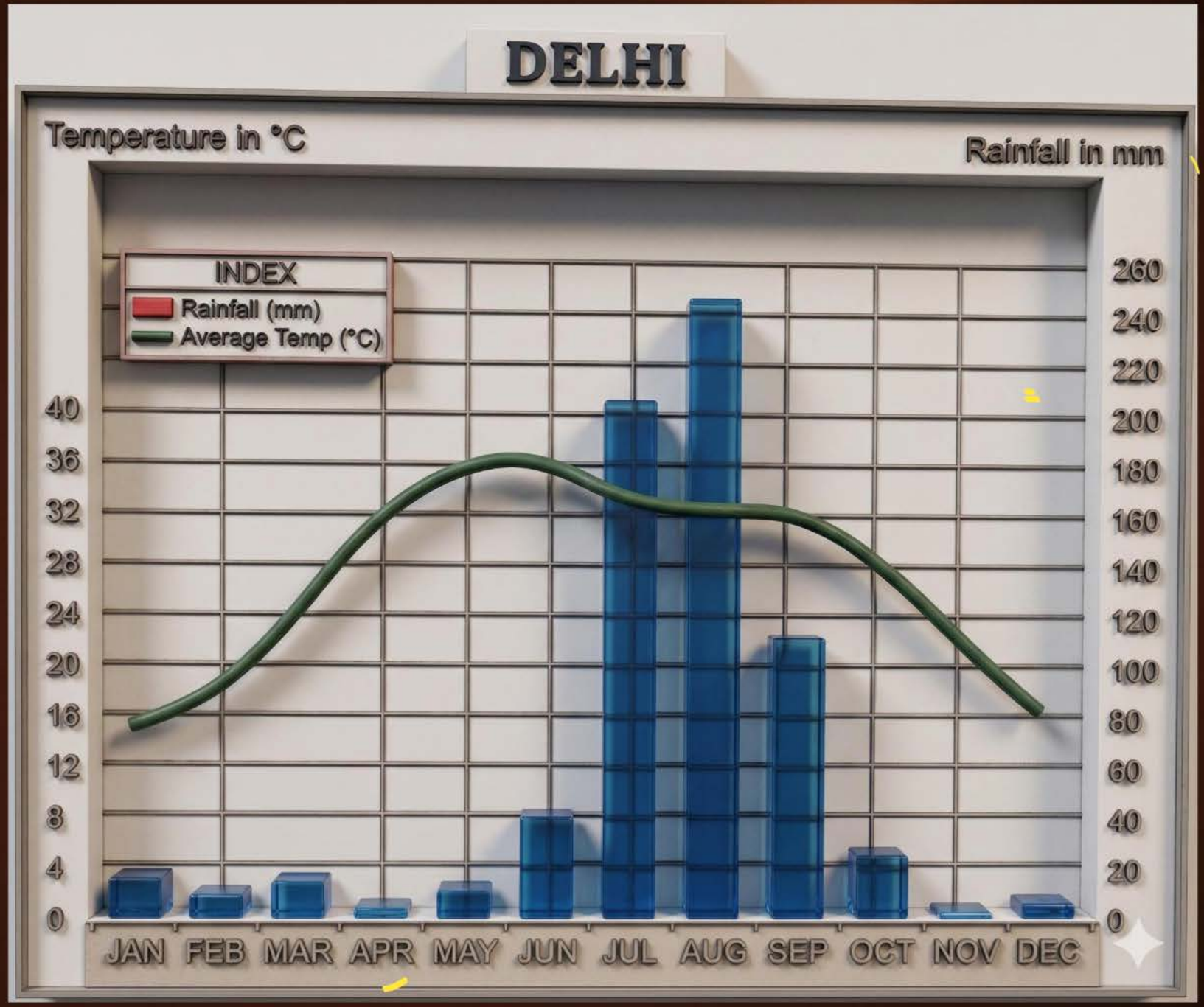

INDIA
Monsoon Winds in September
ARABIAN SEA
BAY OF BENGAL
ANDAMAN & NICOBAR ISLANDS (INDIA)
LAKSHADWEEP (INDIA)
INDIAN OCEAN
Km 0 100 400 600 Km
Scale 1:36,000,000
INDIA
Monsoon Winds in November
ARABIAN SEA
BAY OF BENGAL
ANDAMAN & NICOBAR ISLANDS (INDIA)
LAKSHADWEEP (INDIA)
INDIAN OCEAN
Km 0 100 400 600 Km
Scale 1:36,000,000

India – Winter Rainfall

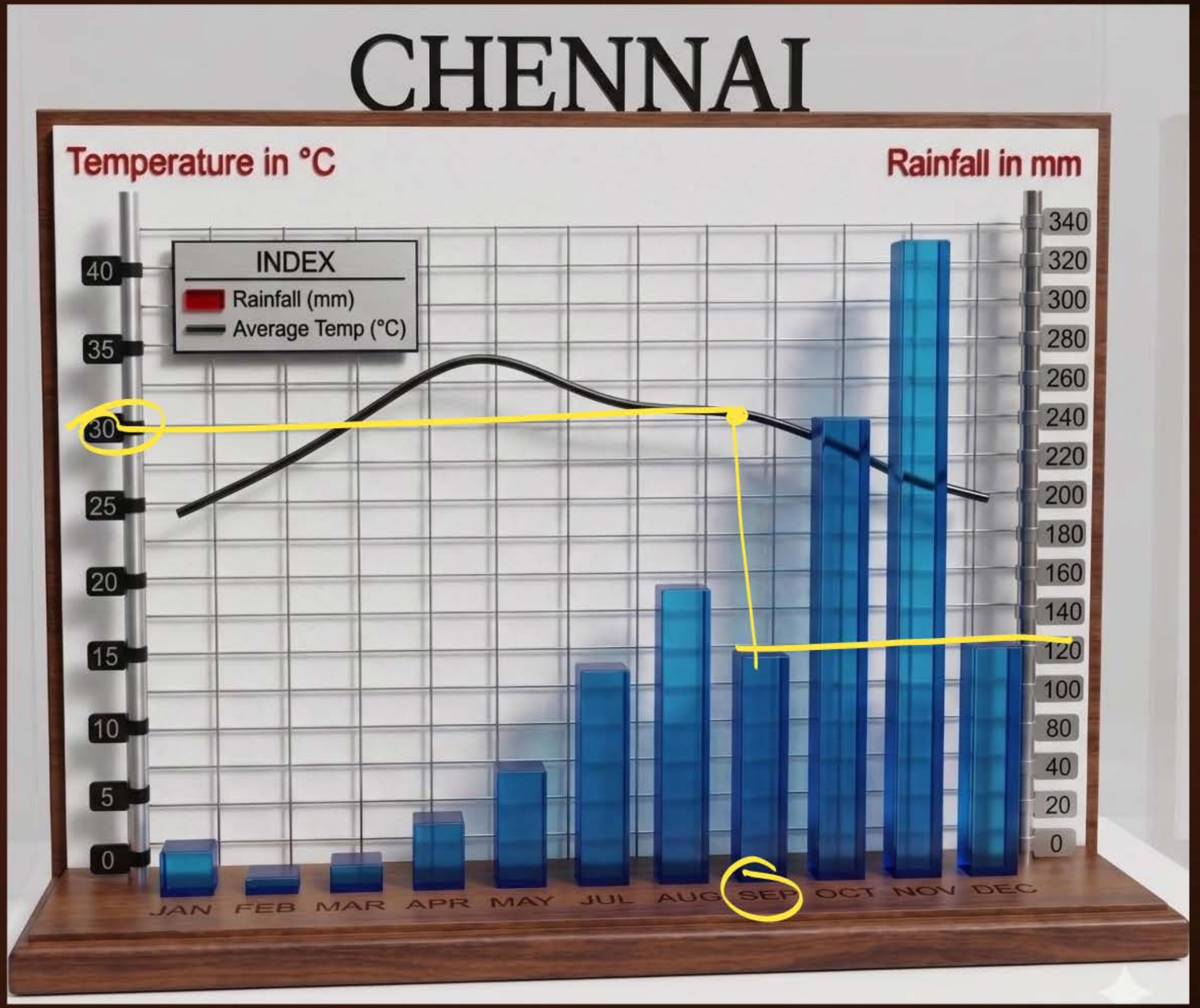


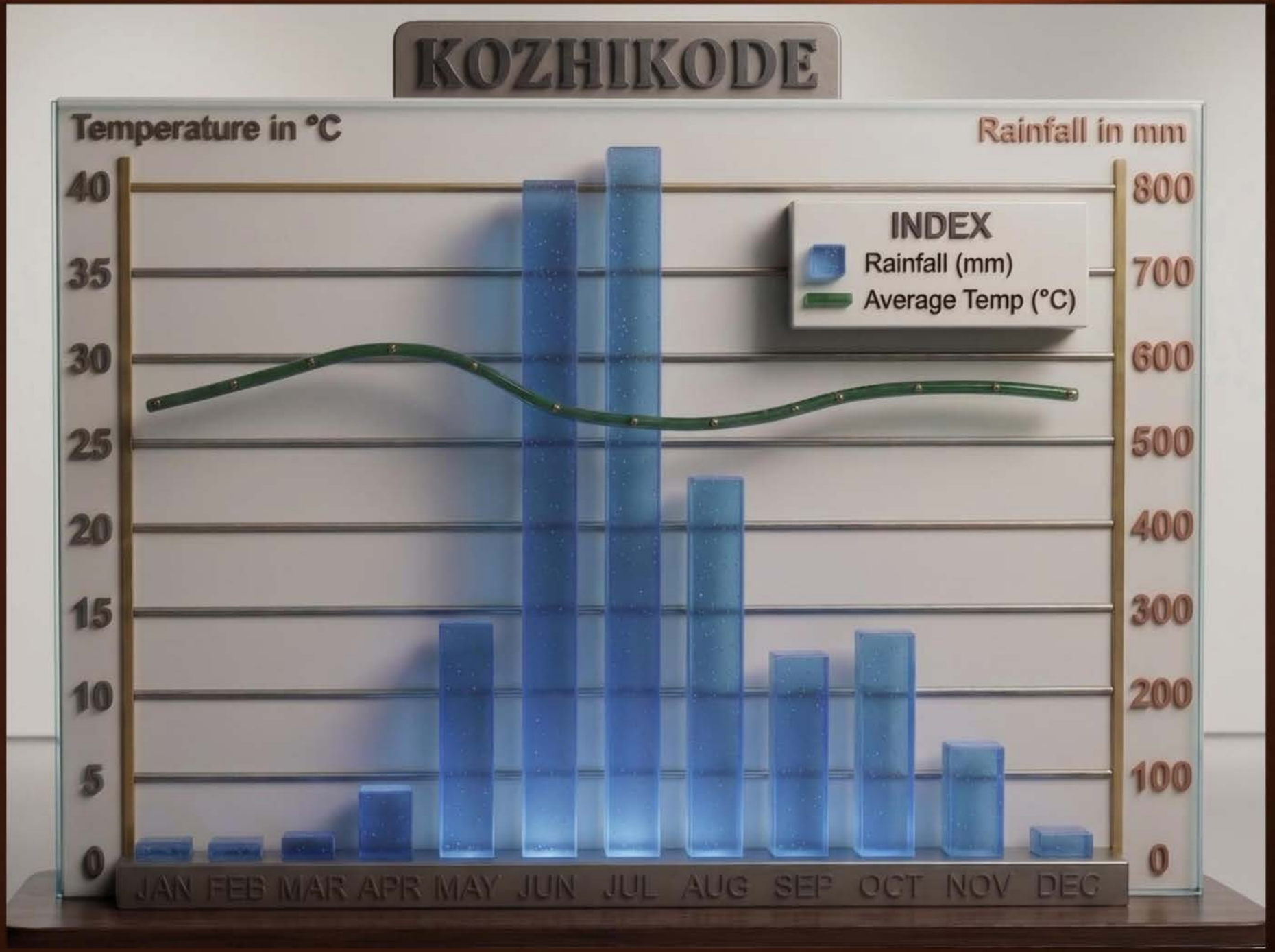
Interpretation of Climatic Data

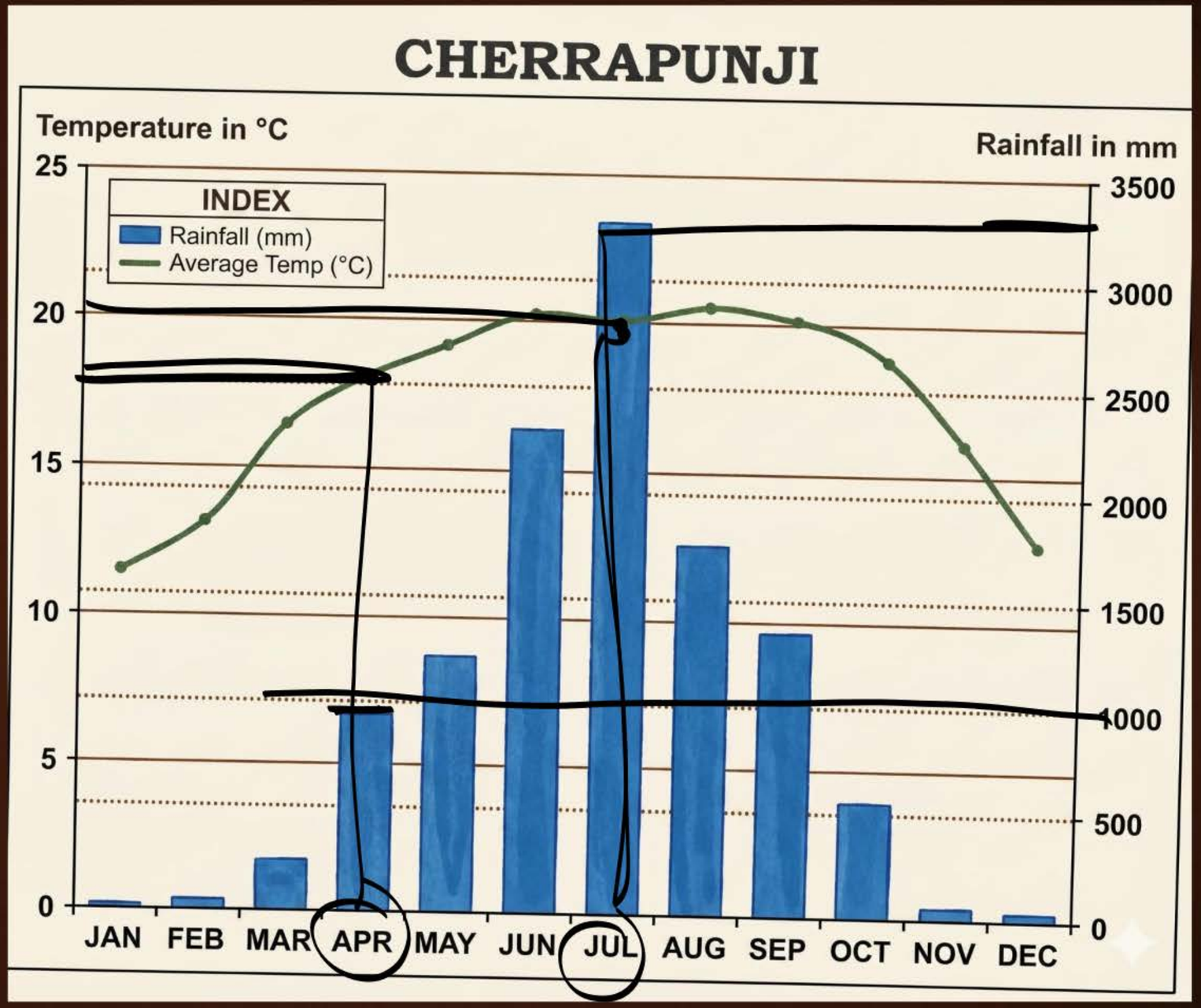



DELHI
Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
40
36
32
28
24
20
16
12
8
4
0
260
240
220
200
180
160
140
120
100
80
60
40
20
0
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC

CHENNAI


Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
40
35
30
25
20
15
10
5
0
340
320
300
280
260
240
220
200
180
160
140
120
100
80
40
20
0
JAN
FEB
MAR
APR
MAY
JUL
AUG
SEP
OCT
NOV
DEC


KOZHIKODE
Temperature in °C
Rainfall in mm
40
35
30
25
20
15
10
5
0
800
700
600
500
400
300
200
100
0
INDEX
Rainfall (mm)
Average Temp (°C)
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC


CHERRAPUNJI
Temperature in °C
Rainfall in mm
INDEX
Rainfall (mm)
Average Temp (°C)
25
20
15
10
5
0
3500
3000
2500
2000
1500
1000
500
0
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
DEC

QUESTIONS

(a) Name the type of climate prevailing over India. Mention any two factors responsible for it.

(b) State two important characteristic features of the Monsoon rainfall in India.

(c) Give a geographical reason for each of the following:
- (i) Heatwaves in the end of April are experienced in the northern plains.
- (ii) Tropical cyclones are common in the Bay of Bengal as compared to the Arabian Sea.
- (iii) The Indo-Gangetic Plain gets some rainfall in the months of December and January.

(d) (i) What is 'Monsoon'?
- (ii) Name the place in India which receives the heaviest rainfall.

QUESTIONS

(a) Mention the different sources of rain in Punjab and Tamil Nadu during the winter season.

(b) What is meant by 'rain-shadow area'? Give an example and state the mountains which are responsible for the rain-shadow area.

QUESTIONS

(c) Study the climatic data given below and answer the questions that follow:

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Temperature °C	23.1	24.8	26.5	29.3	32	32.8	33.1	32.1	30.5	29.3	28.7	26.1
Rainfall cm	15.3	10.1	0.3	0.1	1.3	4.5	6.1	10.2	10.5	20.1	16.8	19.0

(i) Calculate the annual rainfall experienced by the station.

(ii) What can you conclude about the climate of this station - is it maritime or continental? How did you come to this conclusion?

(iii) Name the season during which rainfall is the heaviest. Name a probable station satisfying this criteria.

QUESTIONS

(d) Give a geographical reason for each of the following:

(i) Western Rajasthan receives no rain from the Arabian Sea branch of the South-West Monsoon winds.

(ii) Mangalore is not cold even in the month of December.

(iii) Even in summer Shimla is cooler than Delhi.

Aravalli Hills are parallel.

→ maritime influence

Altitude

QUESTIONS

(a) How does El-Nino affect the climate of India?

(b) State two characteristic features of North-East Monsoon.

(c) Mention three important features of rainfall in India.

(d) Give a geographical reason for each of the following:
- (i) Pune receives less rainfall than Mumbai.
- (ii) Tropical cyclones bring heavy rainfall but temperate cyclones bring light rainfall.
- (iii) Kerala has a tropical climate while Bihar has a continental climate.

QUESTIONS

(a) Give two points of difference between the Arabian Sea branch and the Bay of Bengal branch of South-West Monsoon Winds.

(b) Name any two states that receive rain in January-February from North-East Monsoons.

(c) Give a geographical reason for each of the following:
- (i) Western part of Rajasthan is a desert region.
- (ii) Kochi has a lower annual range of temperature than Agra.
- (iii) The Northern plains and Punjab receive winter cyclonic rain in winter.

(d) (i) What causes the winter rain along Coromandel coast?
- (ii) Name a state which gets sufficient rainfall from the winter monsoon.

QUESTIONS

(a) Give two points of difference between Tropical Cyclones and Temperate Cyclones.

(b) An Inter-School Football Match at Visakhapatnam from November 1 to 3 had to be cancelled because of bad weather. Give an explanation for this happening. Your explanation should provide the technical name of this season in India, as well as the pressure conditions over the Bay of Bengal.

(c) Give a geographical reason for each of the following:
- (i) Kanyakumari experiences an equable climate.
- (ii) The Coromandel Coast remains dry in July-August.
- (iii) Mangalore experiences more rainfall than Mysore.

(d)
- (i) Name one region which gets rainfall from the Retreating Monsoon.
- (ii) Name two areas which get more than 200 cm of rainfall in India.

QUESTIONS

(a) From where does the winter showers in Western Uttar Pradesh and Punjab originate?

(b) State the economic importance of Kalbaisakhis in West Bengal and Assam.

(c) Give a geographical reason for each of the following:
- (i) Chennai receives less rain than Thiruvananthapuram although it has more rainy days.
- (ii) Shillong gets less than 200 cm of rainfall in a year while Cherrapunji receives more than 1250 cm of rainfall.
- (iii) Jaipur has a higher annual range of temperature than Goa.

QUESTIONS

(d) Briefly explain the role of the Himalayas in determining the climate of India during:

(i) the hot weather season;

(ii) the southwest monsoon season;

QUESTIONS

(a) According to a news report the onset of southwest monsoon over Kerala was delayed and happened on June 8, 2023 due to El Nino. What is El Nino? What is the expected date for the onset of Southwest Monsoons over Kerala?

(b) Uneven monsoon with below average rainfall in June, above average rainfall in July and a dry August was reported one year. How does this pattern of rainfall differ from the regular pattern of rainfall over the subcontinent.

(c) Give a geographical reason for each of the following:
- (i) The northern hills of Assam do not get as much rainfall as the southern hills.
- (ii) Northern India has cold-wave conditions in January.
- (iii) The month of October is marked by sultry humid weather.

QUESTIONS

(d) Study the climatic data for cities A and B given below and answer the questions that follow:

City A

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Temperature °C	15.4	19.5	24.9	29.6	31.5	31.7	30	29.8	29.2	27.1	22.3	17.2
Rainfall mm	11	14	10	13	45	165	355	280	200	50	7	6

City B

Month	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Temperature °C	24	26	29	30.8	32	31.6	30	29.6	29.6	28.6	26.4	24.2
Rainfall mm	10	12	11	20	65	120	130	160	200	210	90	8

(i) Calculate the annual rainfall for both cities.

(ii) Where are the two cities located? Give reasons to support your answer.

QUESTIONS

(a) Taru has lived in Pune for her entire life. Now she has to shift to Ludhiana. As a student of geography what would you tell her to prepare her for life in the new city?

(b) Explain why the South West Monsoon winds enter West Bengal from the south or south-west direction.

(c) How would the monsoon mechanism be affected by the absence of the Great Indian Desert?

(

d) Most parts of India do not receive rainfall in the winter season. Why? Name two regions that receive winter rainfall and identify each region's source.

QUESTIONS

The IMD collates rainfall between June and September as SW monsoon. Answer the following questions.

(a) The southwest monsoon typically accounts for about 70% of India's annual rainfall, making it vital for the country's agrarian economy, which contributes approximately 14% to the national GDP. A good monsoon would boost the economy, how?

(b) By 30 September, 2024 regions like Rajasthan, Gujarat, Tamil Nadu, and Goa have recorded extraordinary surpluses, with Rajasthan alone seeing a 57% increase over average rainfall. Would the surplus rainfall be beneficial? How?

QUESTIONS

(c) There have been regional imbalances in the rainfall received with several states having deficits: Manipur (-30%), Bihar (-26%), Punjab (−23%), and Himachal Pradesh (-21%). While the surplus is beneficial for India's overall water availability and agricultural output, states experiencing deficits may continue to face difficulties. With the help of an example explain what difficulties would these states face.

(d) Study the picture on the right. Briefly explain what could be the cause of this calamity and what could be done to bring relief.

Flood



QUESTIONS

Western Disturbances

(a) On my visit to the queen of the hills, the city of Nainital in the month of December, there was snowfall and I stayed indoors. I was thrilled but also amazed as to why it occurred when it was cold winters.

(i) What is the cause for the precipitation to occur at Nainital?

(ii) Explain the phenomenon which leads to the given situation.

(b) At Lucknow, during the month of May, the hot dry season is characterized by a local wind, which increases the temperature for the city and keeps all the people mostly indoors during the afternoon.

(i) What type of local wind is discussed?

(ii) How is it responsible for the given climatic condition prevailing in the above-mentioned city?

North-west → Western Disturbances

Punjab - Haryana → South west → Arabian Sea.

→ South East → Bay of Bengal

Tamil Nadu → Retreating → North East monsoon.

→ Tropical Cyclones

CoCo =/

Inside Out

Toys

=

Cars

Planes

Minions =/

Saturated

Diluted

Juices

Garmi

Thank You